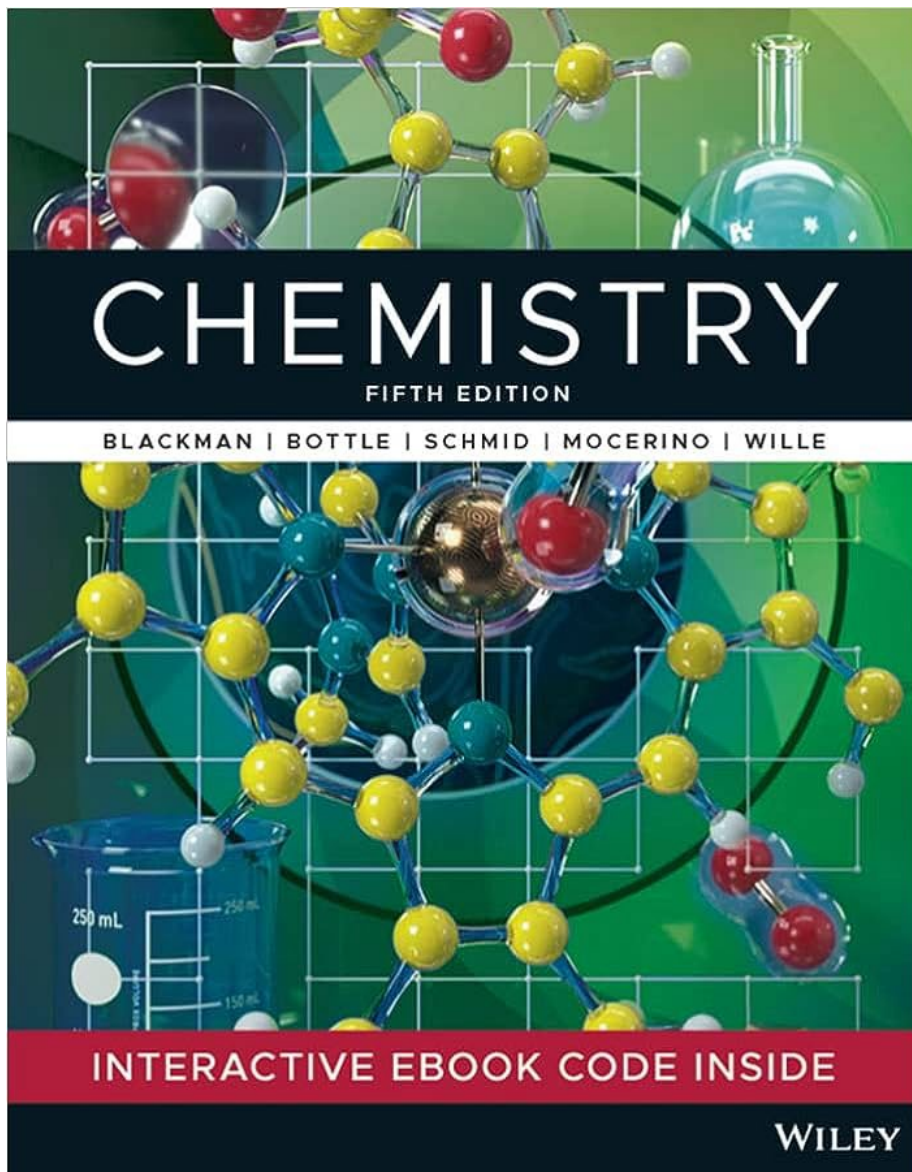




KE559

Kemisk binding

Blackman 5th edition

p. 197-236

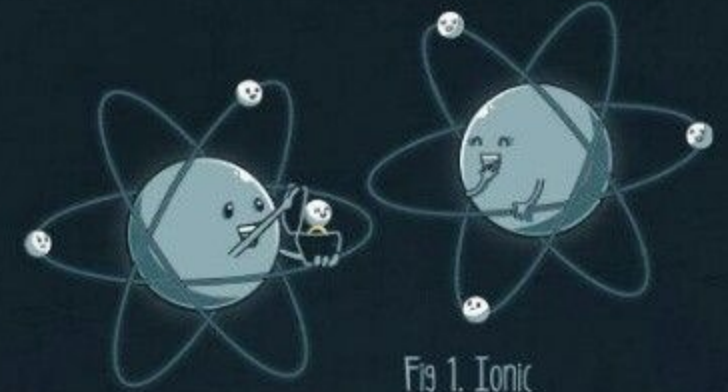
- Kemiske bindinger
 - Typer
 - Kovalente bindinger
 - Sigma- og pi-bindinger
 - Polær-kovalente bindinger
 - Ioniske bindinger
 - Lewis strukturer
 - VSEPR strukturer

Typen af kemiske bindinger

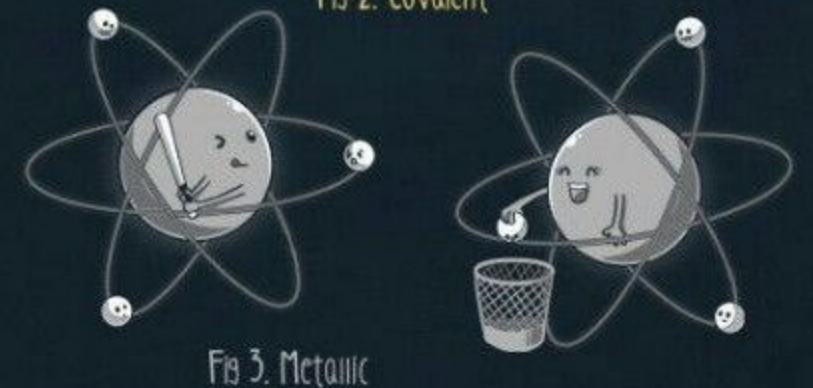
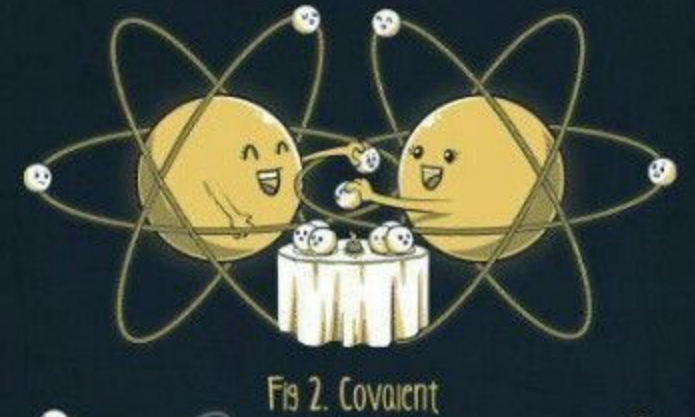
- Kovalent** **Elektroner deles mellem atomer**
- Ionisk** **En eller flere elektroner er overført fra et atom til et andet (iondannelse)**
- Metallisk** **Atomere deler valens elektroner med alle nærmeste naboer**
- Intermolekylære kræfter – van der Waals kræfter:**
Dipole-dipole vekselvirkninger
Hydrogen bindinger
London dispersionskræfter

Bond type	Dissociation energy (kJ/mol)
Ionic lattice	1100-20000
Covalent bond	130-1100
Hydrogen bond	4-50
Dipole-dipole	2-8
London dispersion forces	<4 to 63

⚛ TYPES OF CHEMICAL BONDS ⚛



VIA 9GAG.COM



9GAG is your best source of fun.

Kemiske bindinger

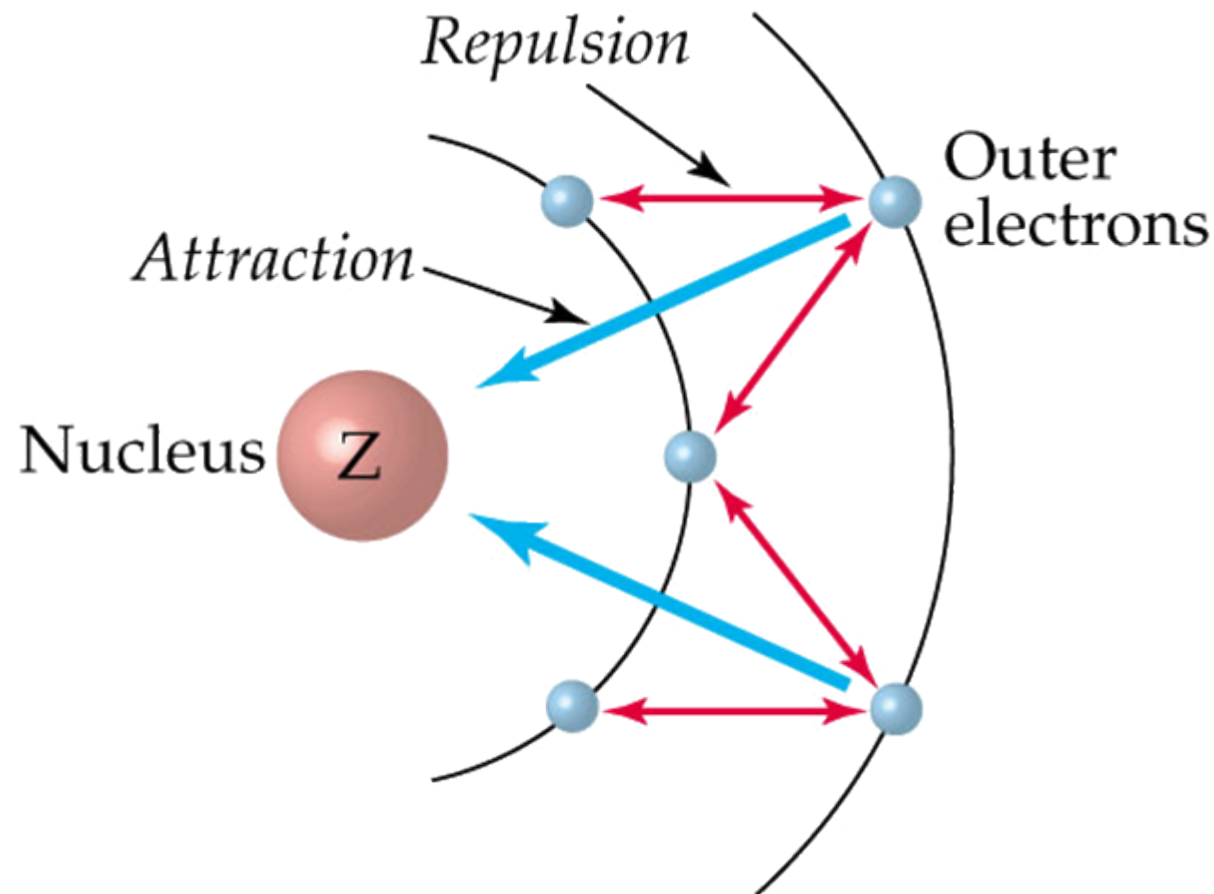
- Opstår som resultat af tiltrækning og frastødning

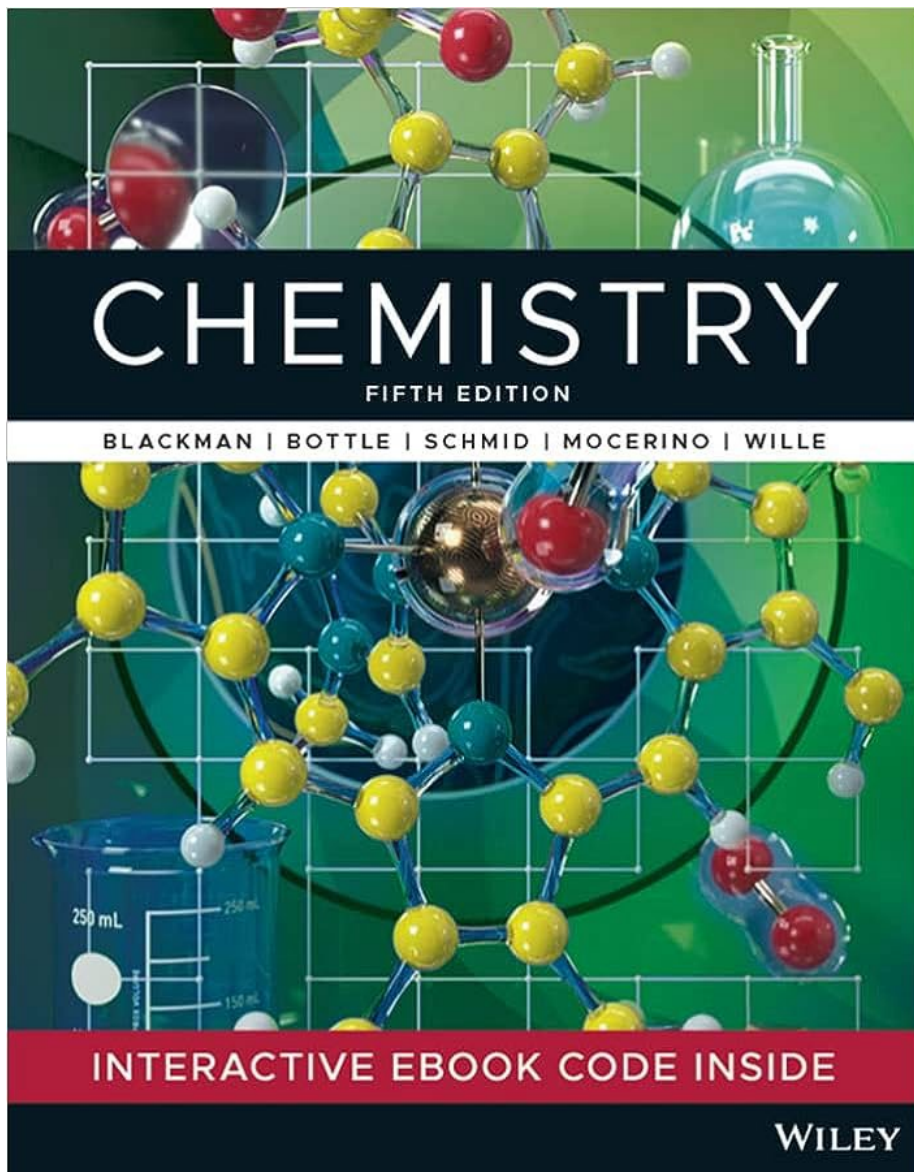
- Positiv <-> Negativ

$$E_{ladninger} = k * \frac{q_1 q_2}{r^2}$$

Coulomb's lov

Coulomb's constant, $k \approx 9 \times 10^9 \text{ N} \cdot \text{m}^2 \cdot \text{C}^{-2}$





KE559

Kemisk binding Ion-binder

Blackman 5th edition

p. 197-236

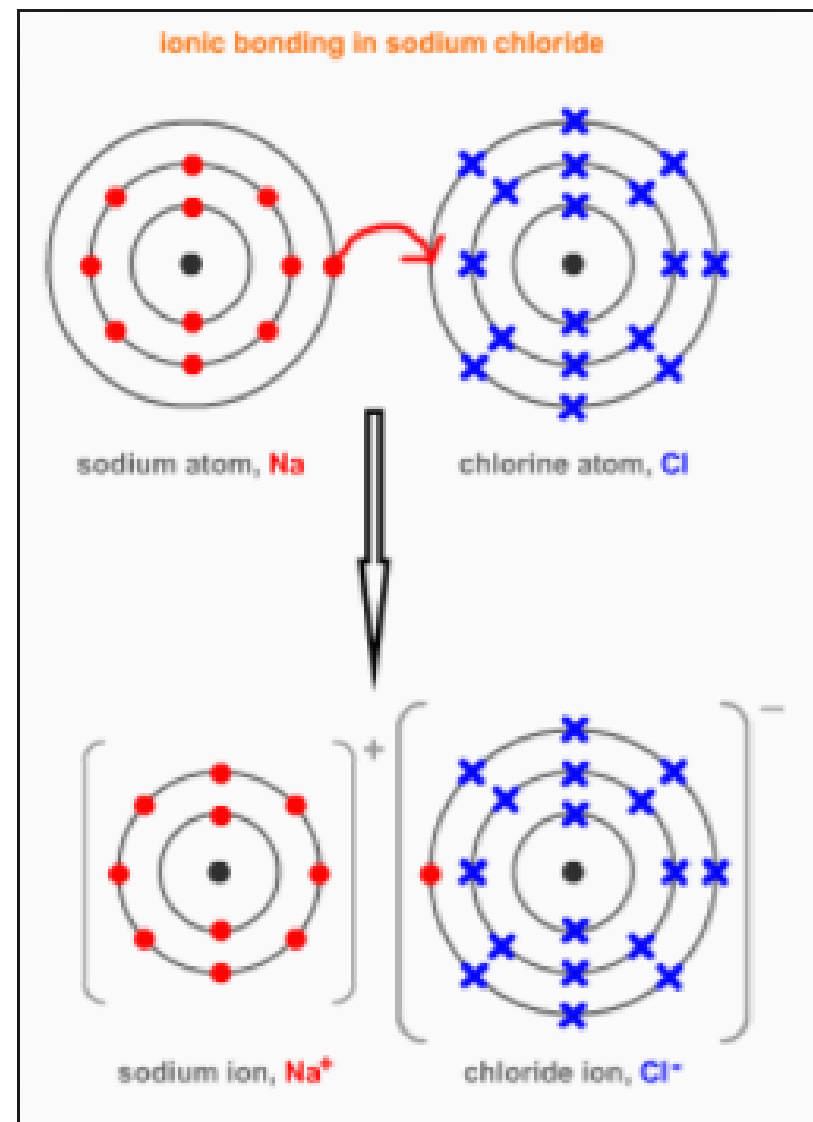
Ionbindinger

- Ioner dannes (som bekendt) ved at en eller flere elektroner overføres fra et atom til et andet atom
- Der dannes en kation og en anion
- Disse har ladning og udøver derfor en elektrostatisk kraft på hinanden

Positiv <-> Negativ

$$E_{\text{ladninger}} = k * \frac{q_1 q_2}{r}$$

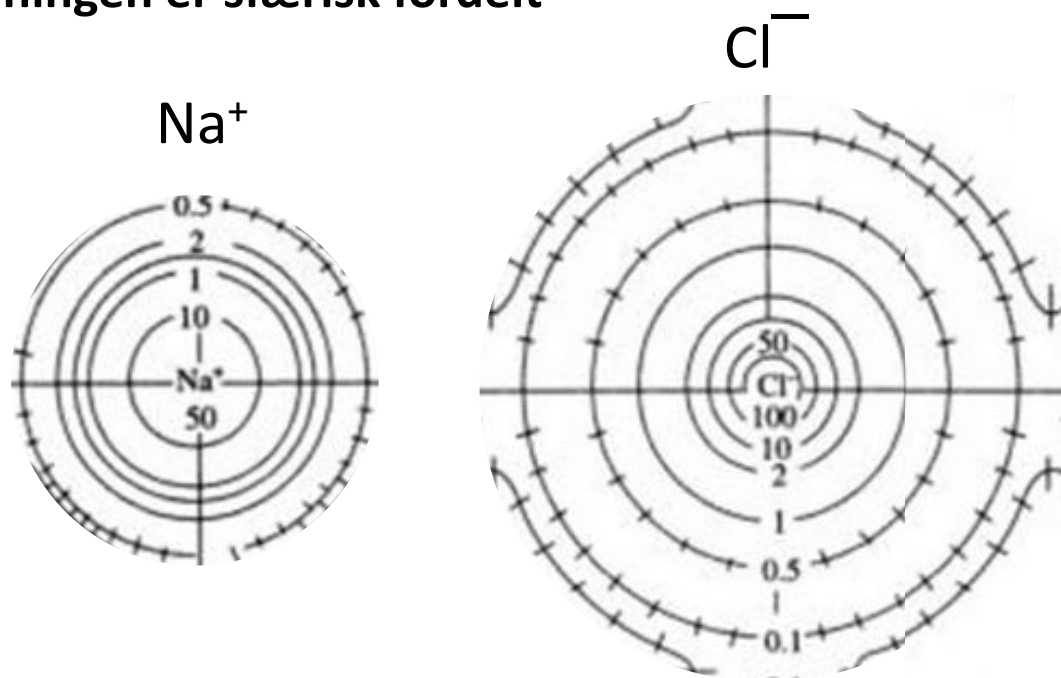
Coulomb's lov



Ionbindinger

Hvordan er ladningen fordelt på en ion? Hvor "sidder" ladningen?

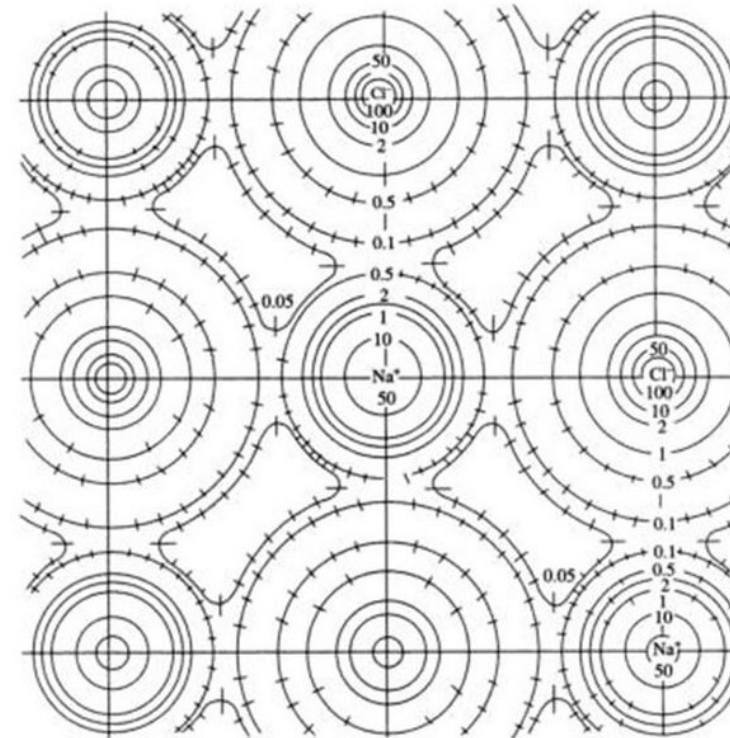
- Ladningen er sfærisk fordelt



- ⇒ Bindingen kan ikke være orienteret i en retning
- ⇒ Ionerne må binde i alle retninger

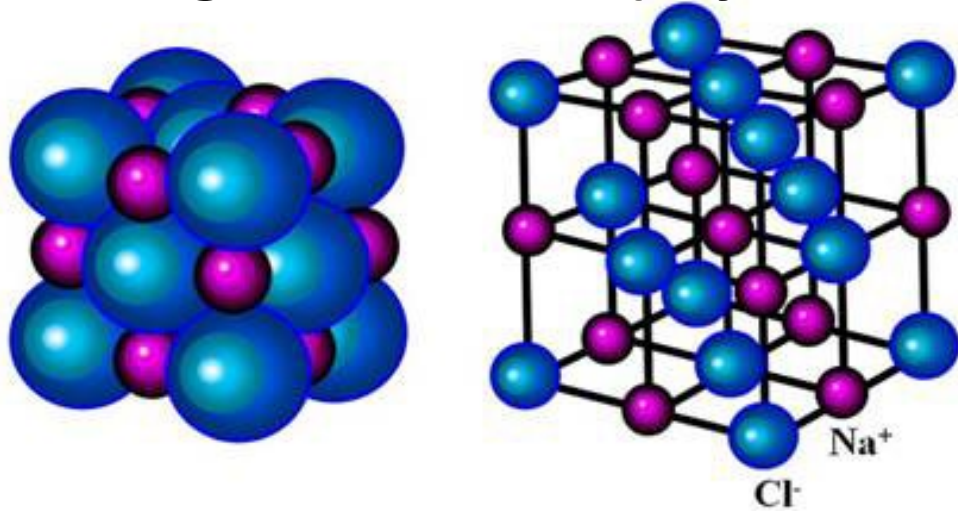
~~Na-Cl~~ molekyle – eksisterer ikke

NaCl
Electron density map

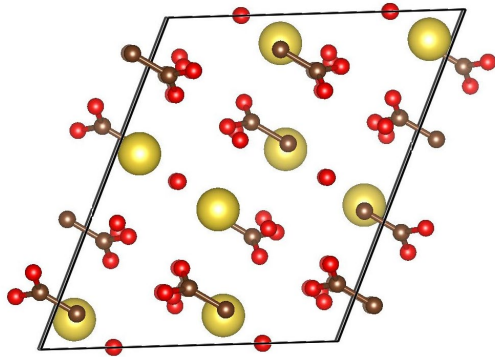


Ionbindinger

NaCl i 3D – gitterstruktur (krystalstruktur)

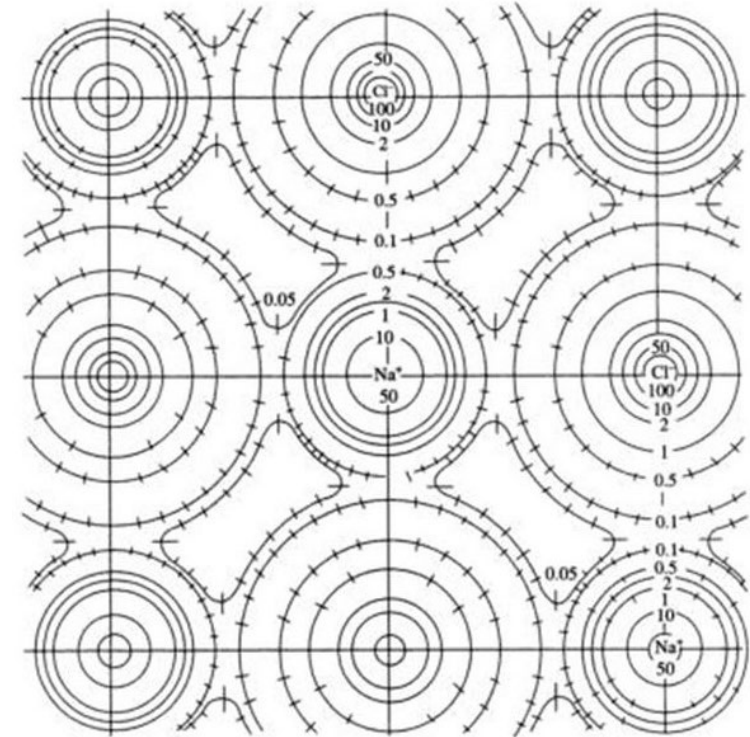


Na-acetat – gitterstruktur (krystalstruktur)



NaCl

Electron density map



Gitterenergi

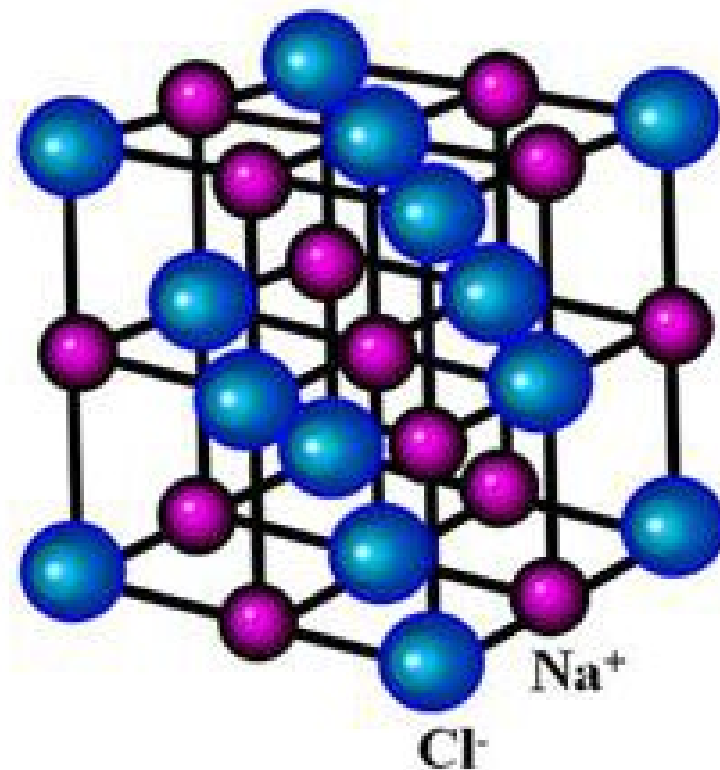
Tages alle tiltrækkende og frastødende kræfter i krystallen i betragtning fås gitterenergien.

Gitterenergi: Den energi det koster at omdanne et mol ionisk stof til de frie ioner på gas form

Positiv <-> Negativ

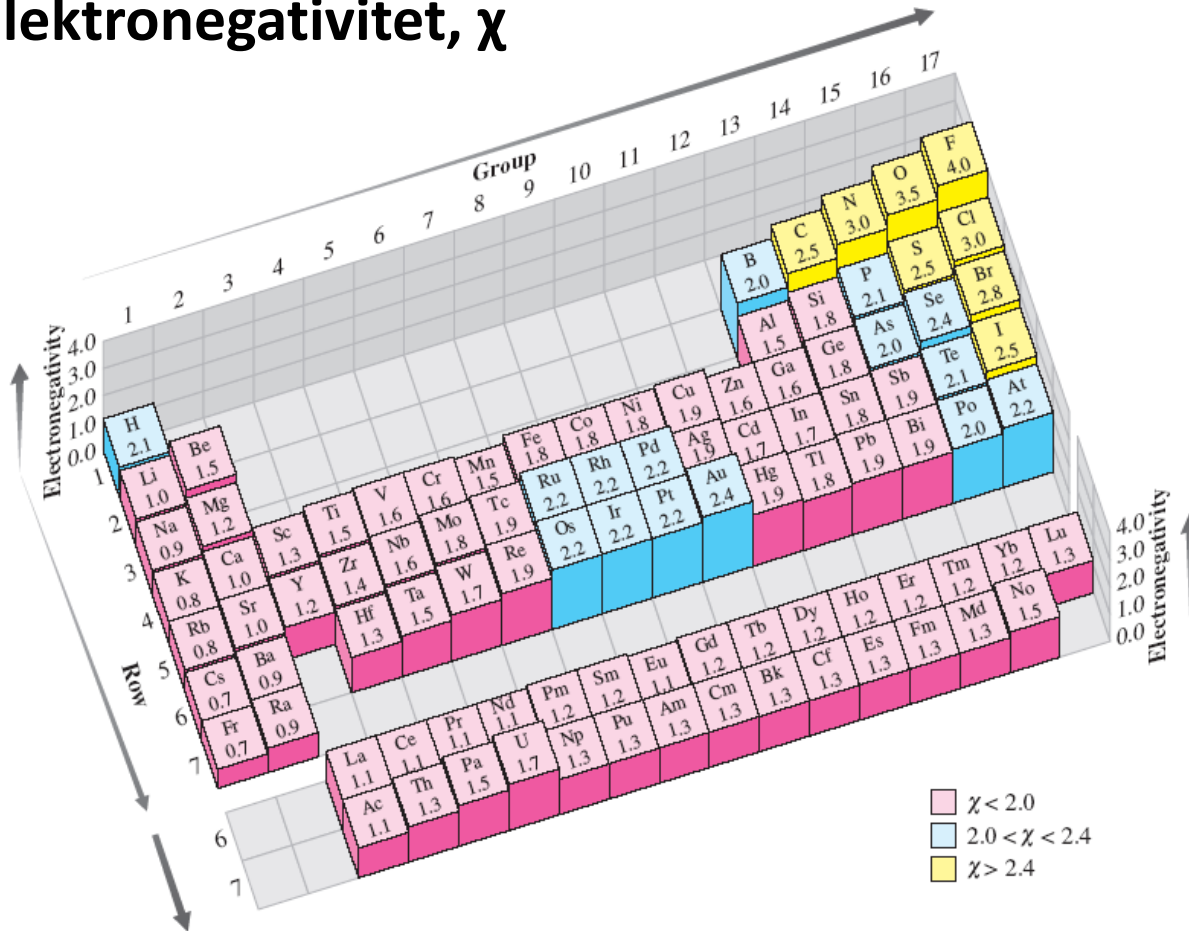
$$E_{ladninger} = k * \frac{q_1 q_2}{r}$$

Coulomb's lov



Hvilken type kemisk binding?

Elektronegativitet, χ



Elektronegativitetsforskel

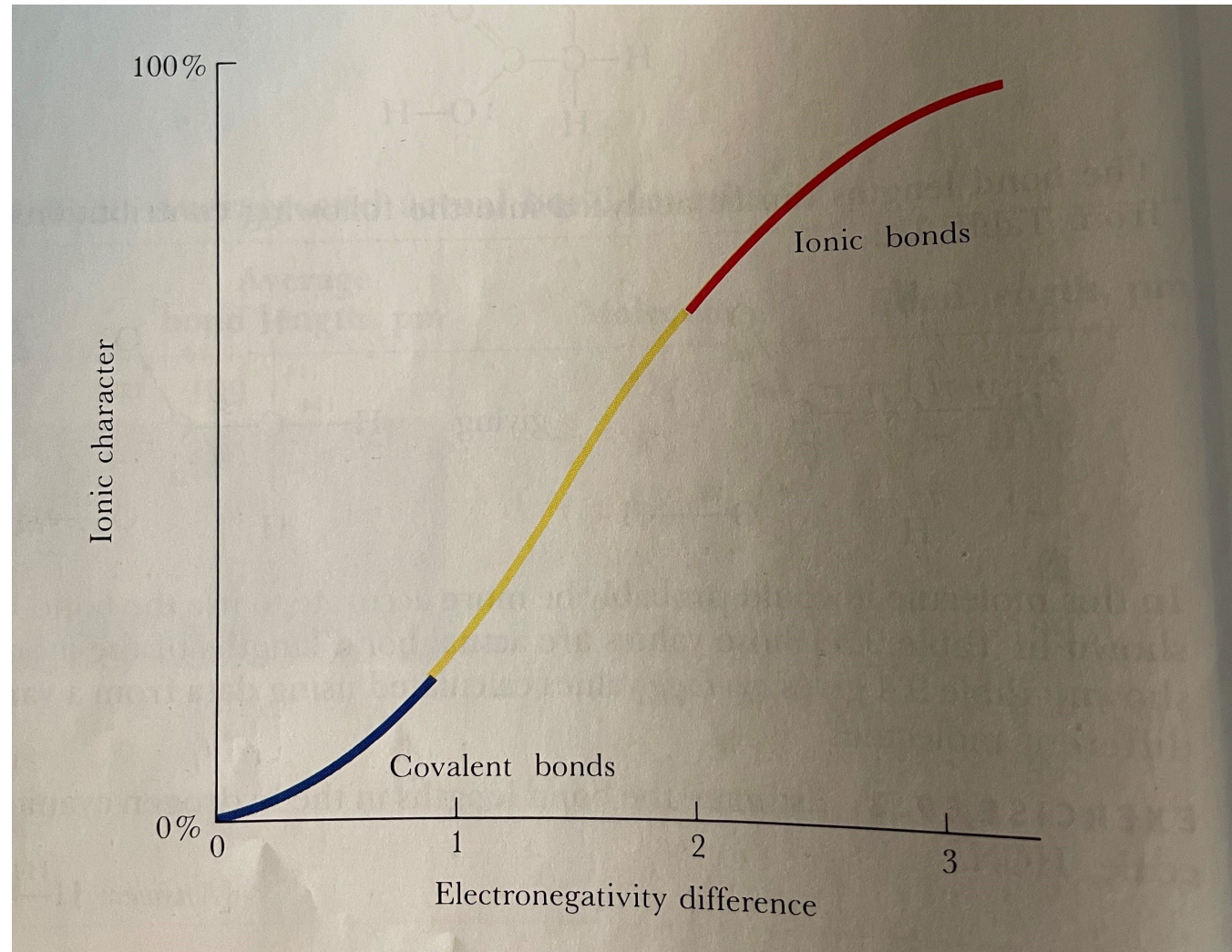
Ionbinding

Polær kovalent binding

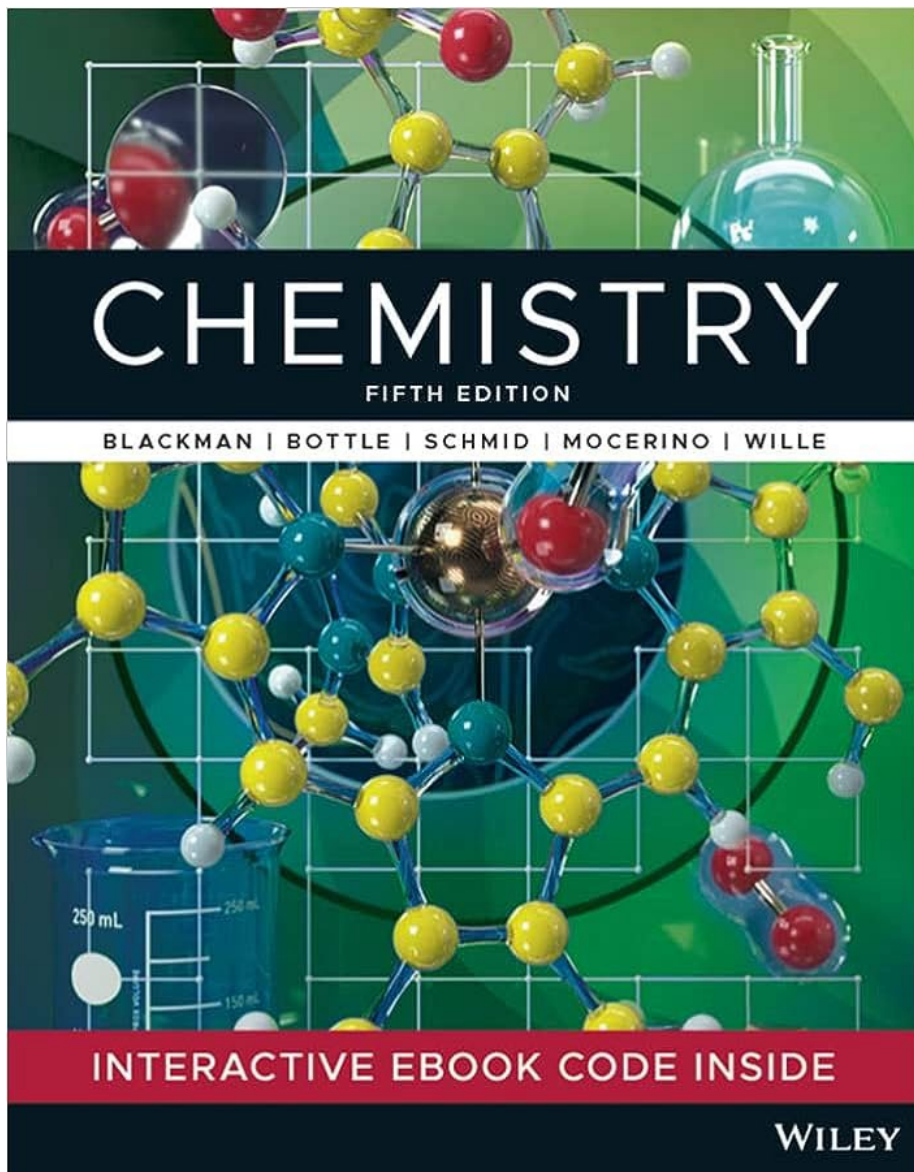
Kovalent binding

Elektronegativitet: Hvor meget vil et atom gerne have en ekstra elektron

Elektronegativitetsforskelle



Billede fra Atkins & Beran, 1990



KE559

Kemisk binding

Kovalente bindinger

Blackman 5th edition

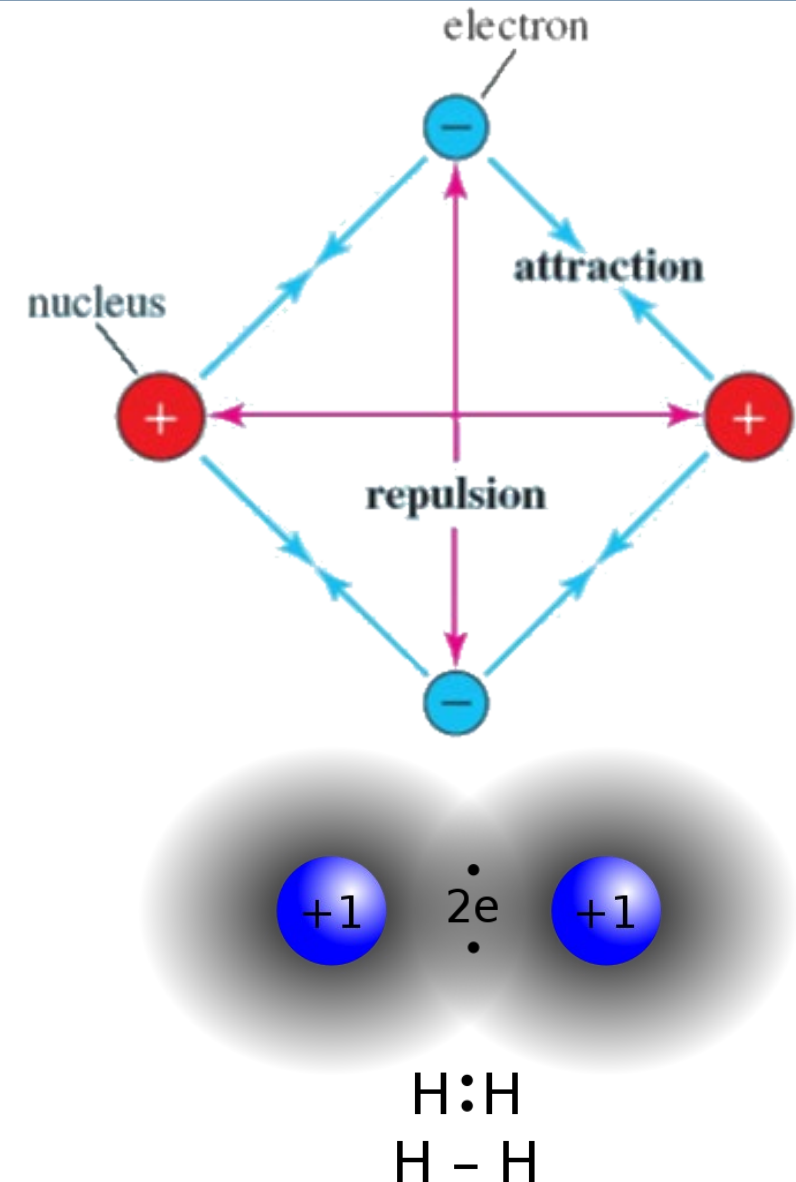
p. 197-236



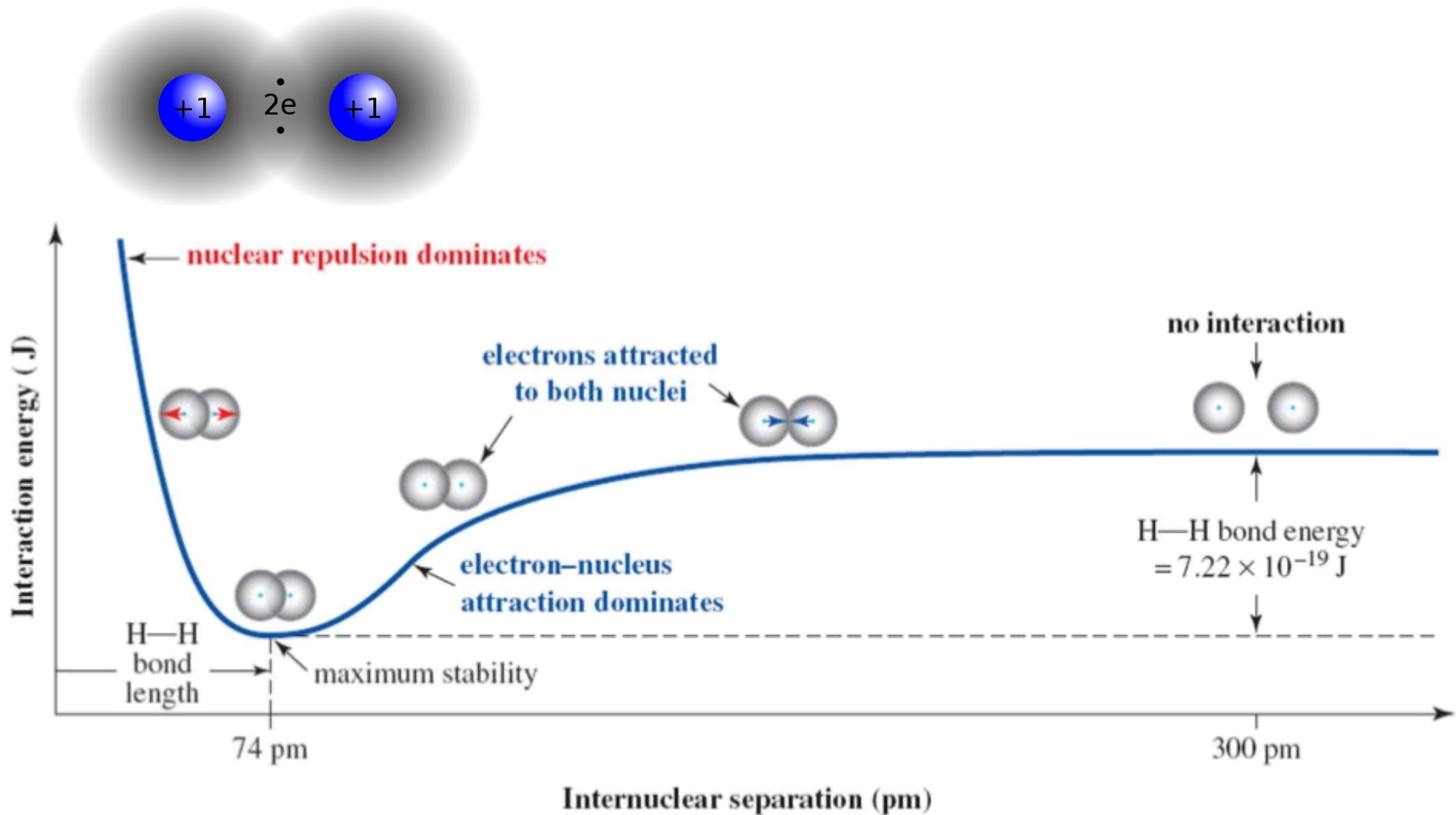
Elektoner deles mellem atomer

Kovalente bindinger

- Positiv <-> Negativ
$$E_{\text{ladninger}} = k * \frac{q_1 q_2}{r}$$
- Balance mellem frastødning og tiltrækning => giver stabilt molekyle med en hvis afstand mellem atomerne
- Overlap mellem atom orbitaler => danner molekylorbital ml. de to atomer
- Molekylorbitalen indeholder to elektroner

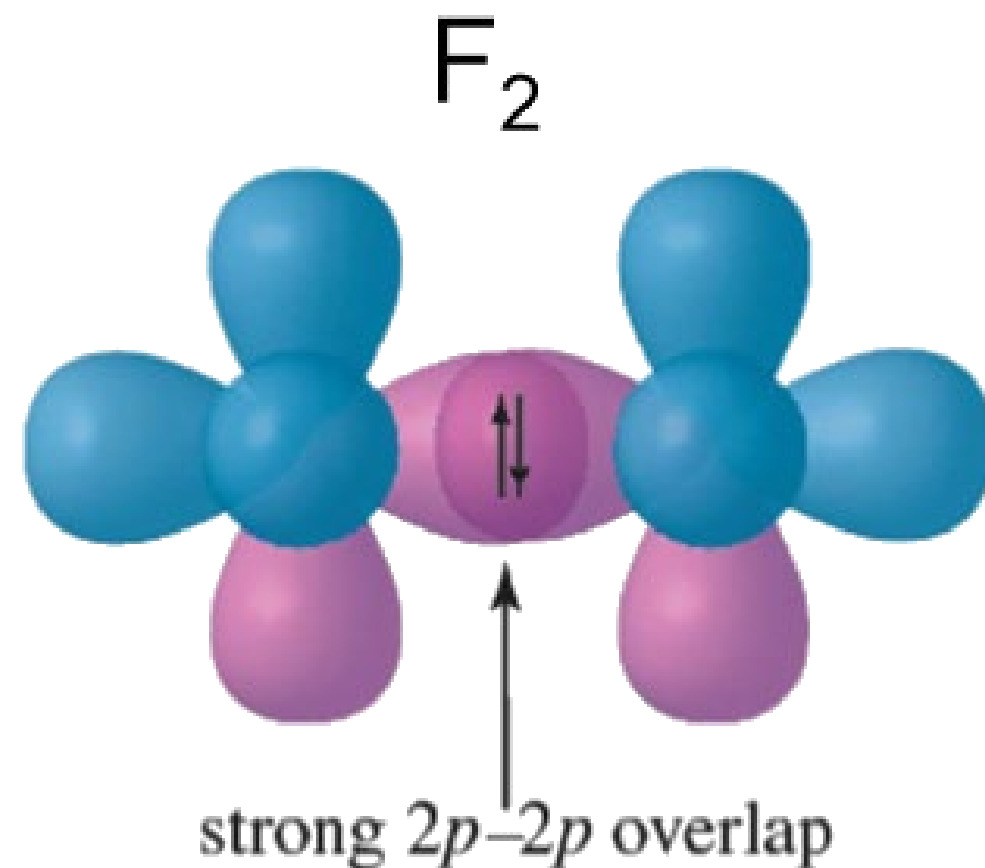


Kovalente binding – Orbital-overlap



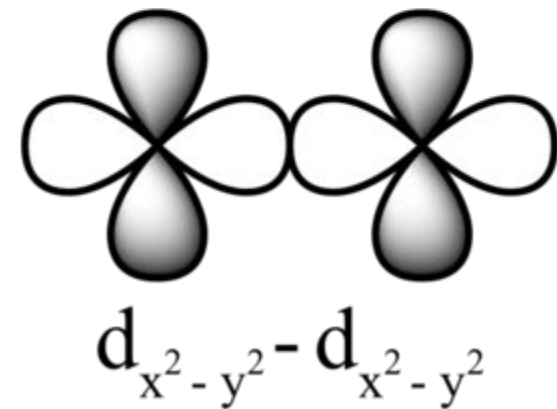
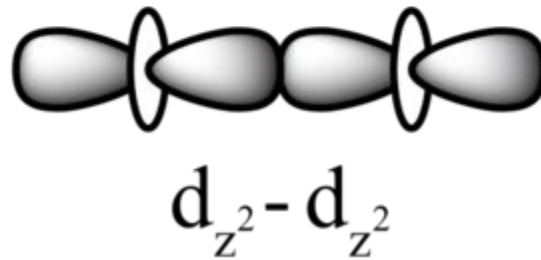
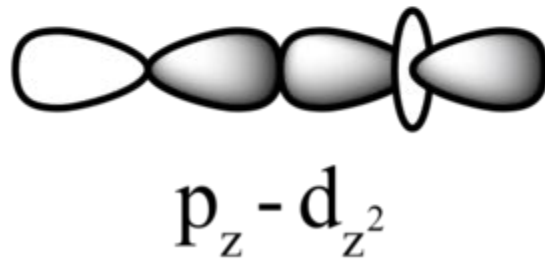
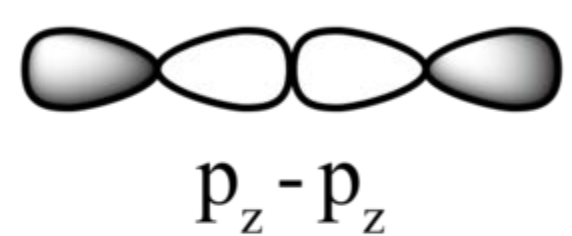
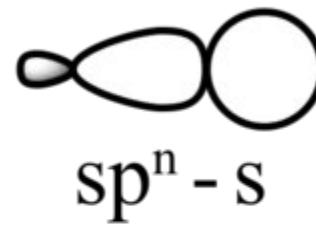
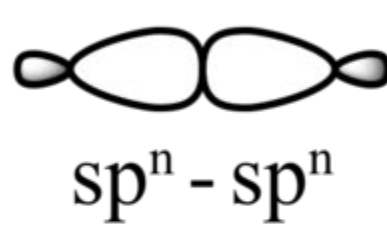
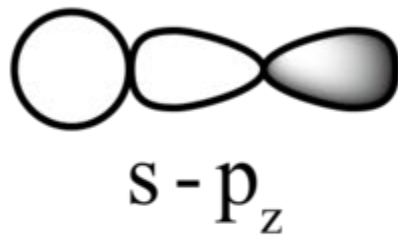
Kovalente bindinger

- F har ni elektroner
- Men kun to elektroner deles i en kemisk binding
- En sigma-binding



Sigma-binder

σ bonds



Hybridisering

- Sigma-binding = overlap, som danner en kovalent binding
– eks I med I til I_2 .
- Hybridisering = hvordan orbitaler blandes i molekylet, eks CH_4

Hybridisering	Geometri	Eksempel	Bindinger
sp^3	Tetraedrisk ($109,5^\circ$)	CH_4 (methan)	4 σ -bindinger (ingen π)
sp^2	Trigonal-plan (120°)	C_2H_4 (ethen)	3 σ -bindinger pr. C + 1 π -binding (fra dobbeltbindingen)
sp	Lineær (180°)	C_2H_2 (ethyn)	2 σ -bindinger pr. C + 2 π -bindinger (fra trippelbindingen)

Hvor mange sigma-bindinger har $\text{CH}_2 = \text{CH}_2$?

a) ☐ 0

b) ☐ 1

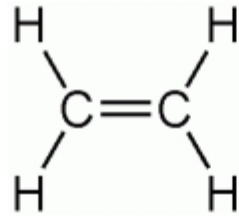
c) ☐ 2

d) ☐ 3

e) ☐ 4

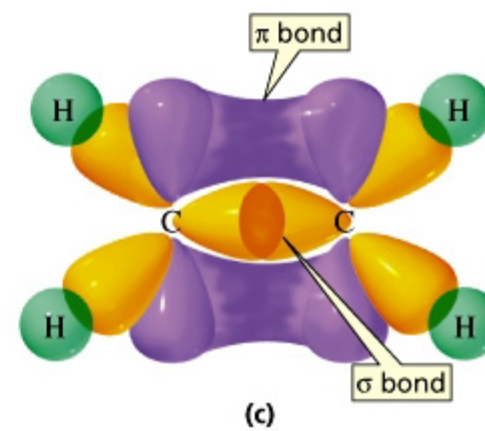
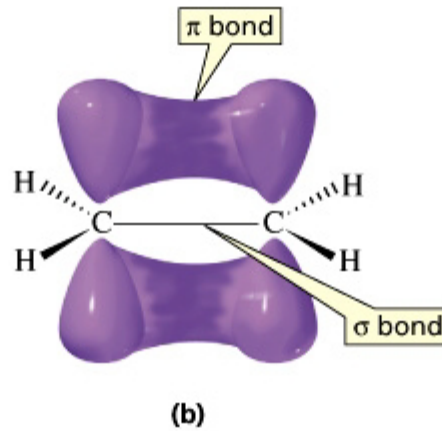
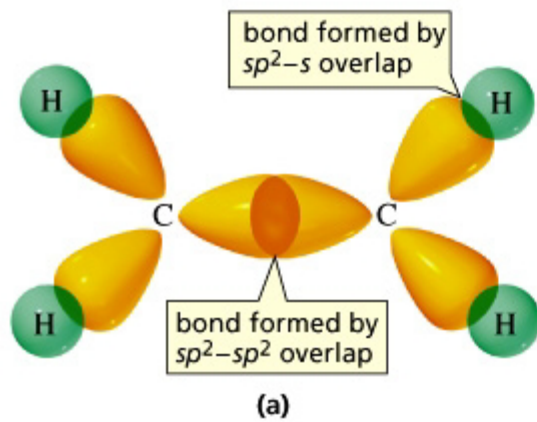
f) ☐ 5

Pi-binding



Ethylene (Ethene)

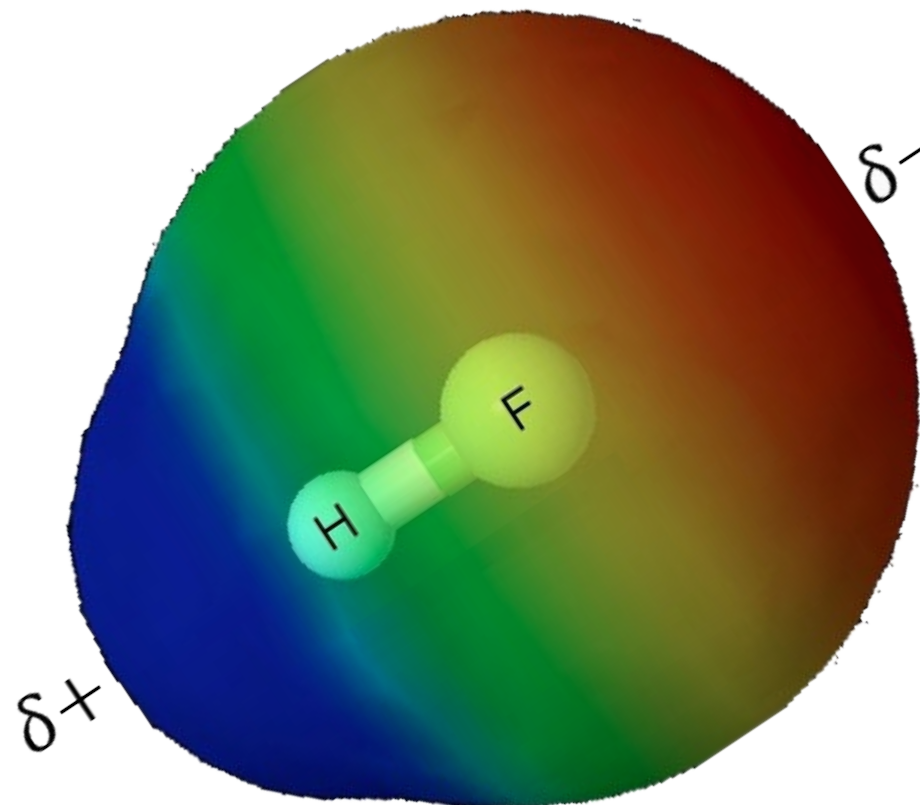
5 sigma
binder



Polær-kovalente bindinger

- Ikke lige deling af elektronen mellem de to atomer
- Dvs. der opstår en delvis negativ og positiv ladning
 - Vi betegner det med $\delta +$ og $\delta -$
 - Mindre end ladninger på ioner som f.eks. Na^+ ladningen

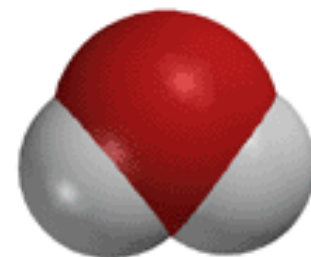
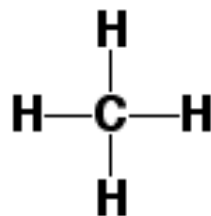
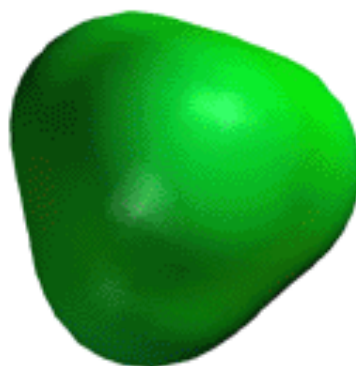
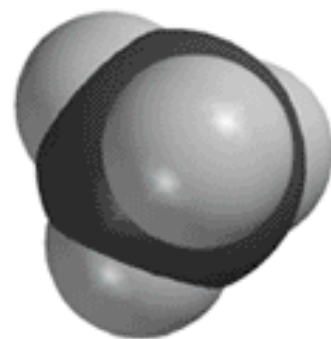
Skyldes elektronegativiteten
 $\chi(\text{H}) = 2.1$, $\chi(\text{F}) = 4.0$



Kovalent vs. Polær-kovalent binding

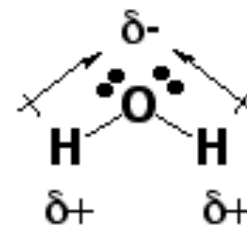
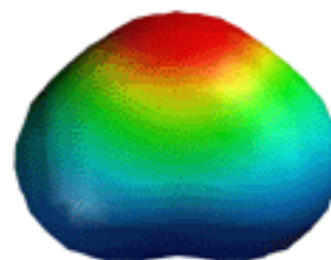
CH_4 – metan:
 $\chi(\text{C}) = 2.5, \chi(\text{H}) = 2.1$

Elektronerne er tæt på
ligeligt fordelt mellem C
og H => ikke-polært
molekyle



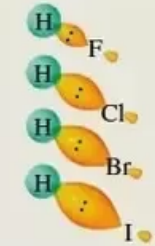
H_2O – Vand:
 $\chi(\text{O}) = 3.5, \chi(\text{H}) = 2.1$

Elektronerne sidder
primært på O => polært
molekyle



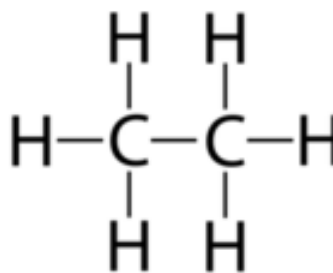
Bindingsenergi og bindingslængde

- Jo større bindingsenergi => des stærkere binding
- Jo kortere bindingslængde => des stærkere binding
- Bindingsenergien stiger når flere elektroner er involveret i bindingen:
 - $E(\text{C}-\text{C})$: ca. 350 kJ/mol ($2 e^-$)
 - $E(\text{C}=\text{C})$: ca. 600 kJ/mol ($4 e^-$)
 - $E(\text{C}\equiv\text{C})$: ca. 840 kJ/mol ($6 e^-$)

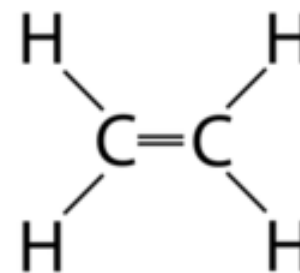
Hydrogen halide		Bond length (Å)		Bond strength	
		kcal/mol	kJ/mol		
H—F		0.917	136	571	
H—Cl		1.2746	103	432	
H—Br		1.4145	87	366	
H—I		1.6090	71	298	

pK_a
3.2
-8.0
-9.0
-10

Ethane C_2H_6



Ethene C_2H_4

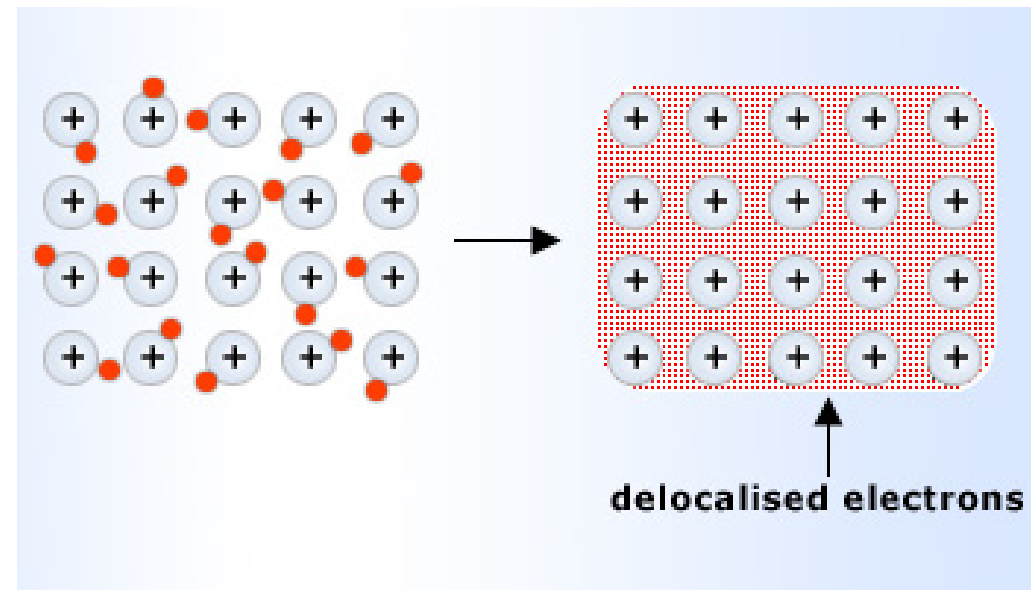


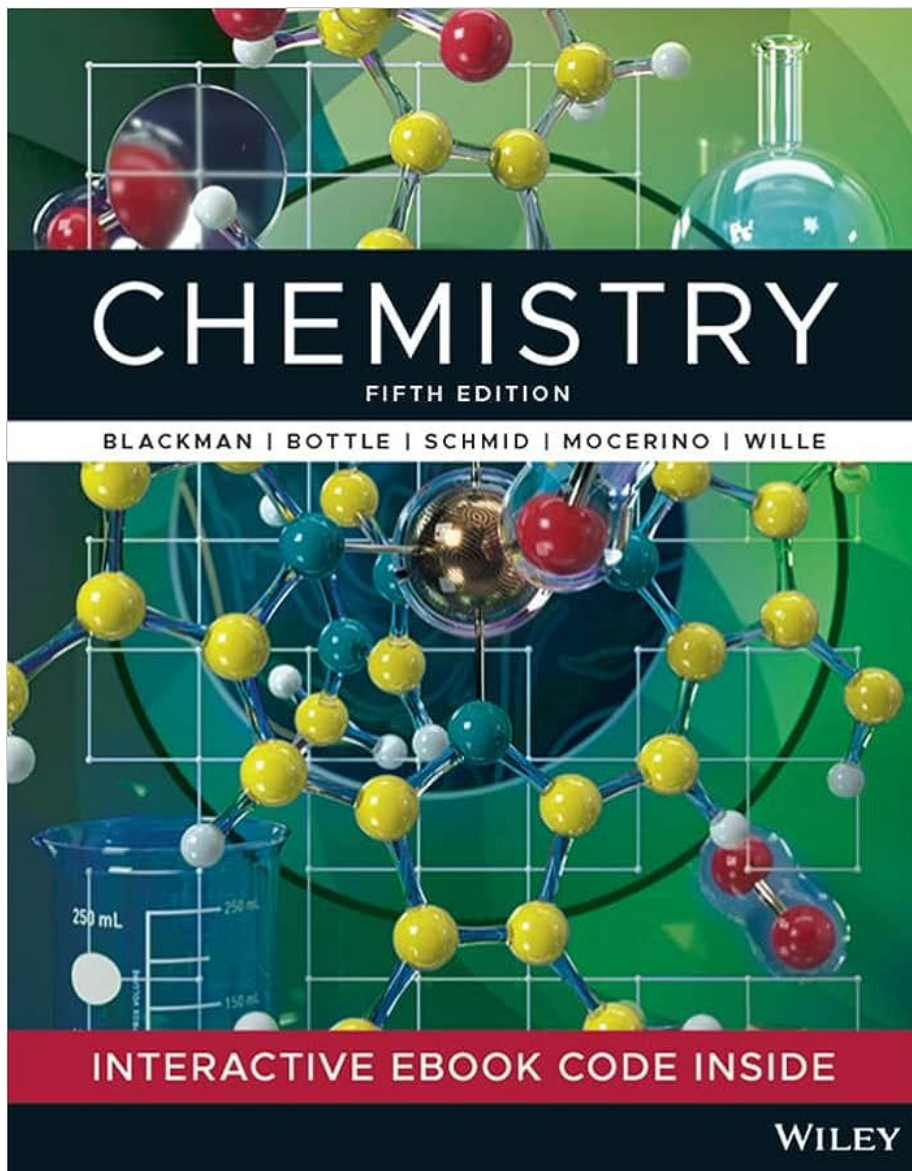
Ethyne C_2H_2



Metallisk binding

- Metal bindinger er anderledes end kovalente eller ioniske bindinger – da elektronerne i metaller er delokaliseret, dvs ikke blot delt mellem to atomer, men flyder frit i et metal gitter
- Dette giver metaller deres særlige egenskaber, inkl. det meget høje smelte punkt

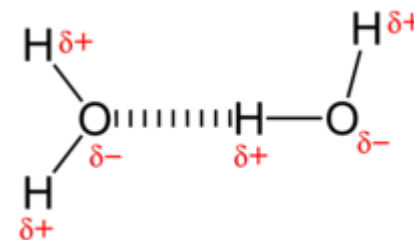


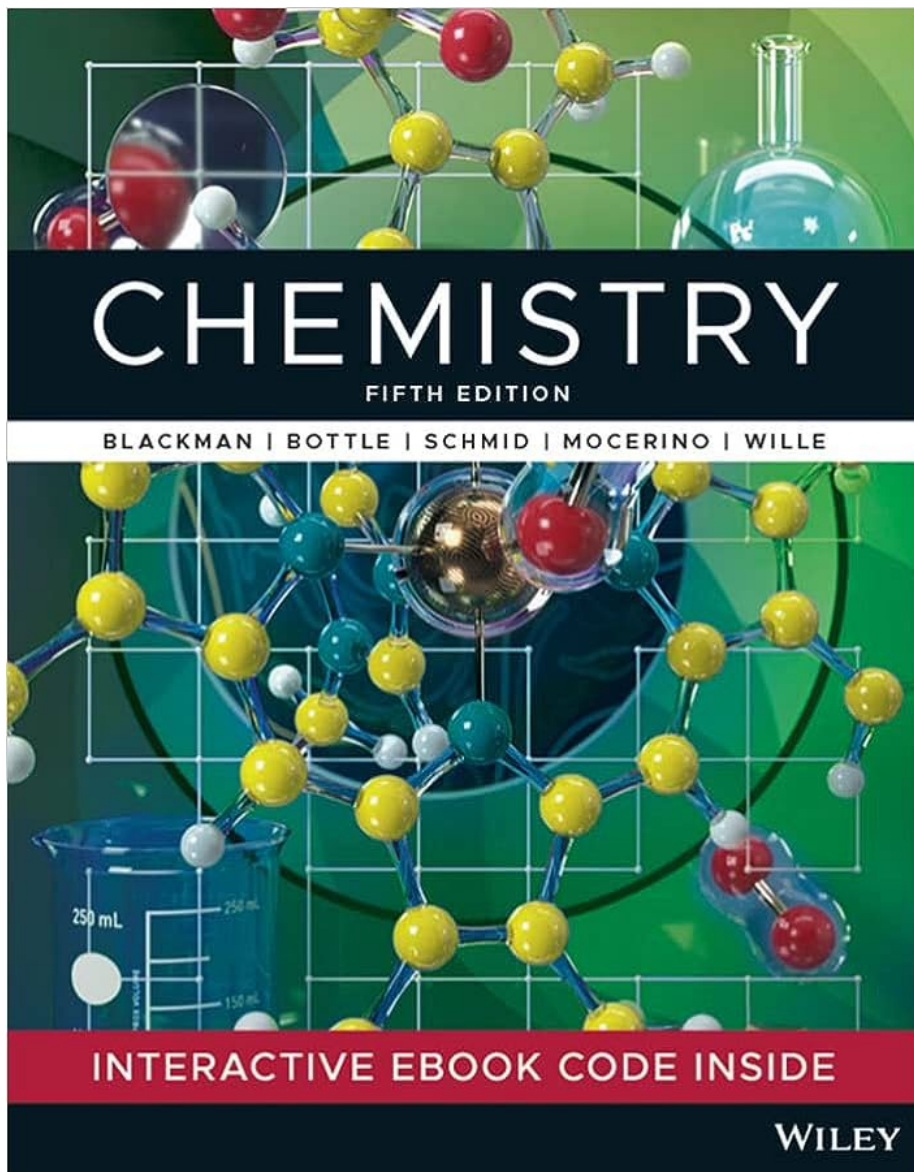


Intermolekylære bindinger

Intermolekylære kræfter

- Hydrogen bindinger
 - Dipol bindinger centeret om hydrogen
- Dipole-dipole vekselvirkninger
 - Samme koncept som ovenfor
- van der Waals kræfter
 - Interaktion mellem uladede molekyler
 - Den stærkeste af van der Waals kræfterne er London dispersionskræfter – som skyldes fluktuationer i elektron placeringen og dermed også i ladningen





KE559

Kemisk binding

Lewis-prik-
strukturer

Blackman 5th edition

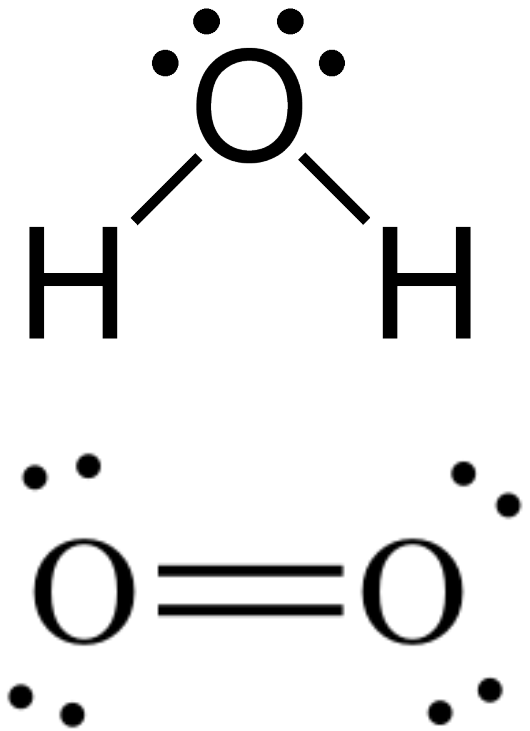
p. 197-236

Lewis struktur

Metode til at forudsige bindingerne i små molekyler

Regler

1. Hvert atom betegnes med symbol
2. Kun valenselektroner indgår
3. En "streg" repræsenterer en kovalentbinding i form af 2 delte elektroner
4. Prikker ved siden af grundstofferne repræsenterer ikke bindende elektroner – oftest forekommer disse i par (lone pairs)

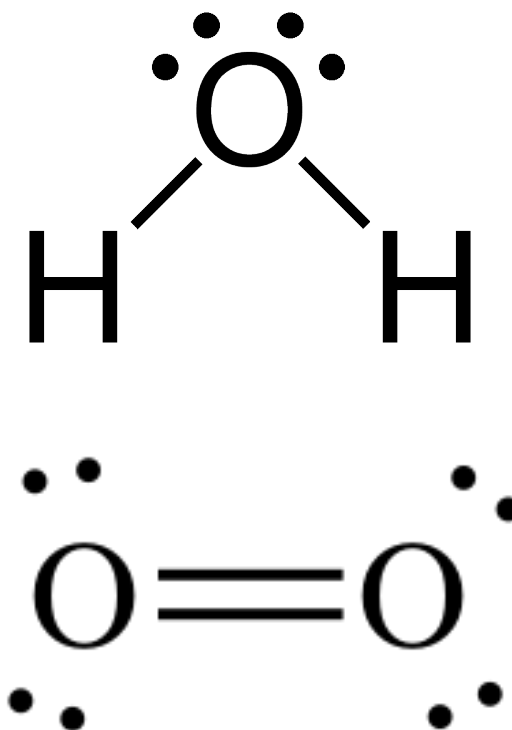


Lewis struktur - fremgangsmåde

Metode til at forudsige bindingerne i små molekyler

Fremgangsmåde (side 205)

1. Tæl hvor mange valenselektroner hele molekylet har (husk at tage højde for evt. ladning)
2. Tegn molekylet med kun enkeltbindinger
3. Placer resten af valens elektronerne ved at placere tre lone-pairs på hvert af de ydre atomer (dog ikke på H)
4. Placer de resterende valens elektroner som lone pairs på det centrale atom
5. Minimer den formelle ladning på alle atomer



Lewis struktur – eksempel CO₂

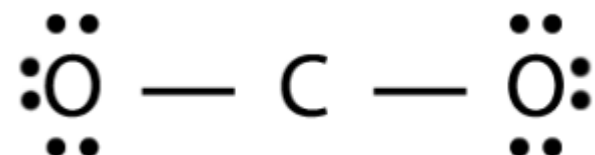
1. Tæl valenselektroner:

$$4 + 6 * (2) = 16$$

2. Tegn strukturen med enkeltbindinger:

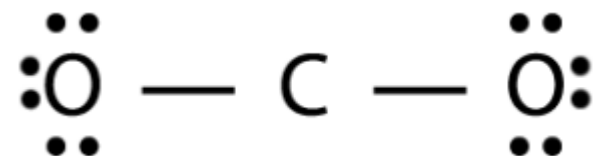


3. Tilføj 3 lonepairs til de ydre atomer (ikke H)



4. Tilføj resterende elektroner til indre atom.

0 tilbage:



Periodic Table of the Elements

Carbon 6 C 12.011

Oxygen 8 O 15.999

Lewis struktur – eksempel CO₂

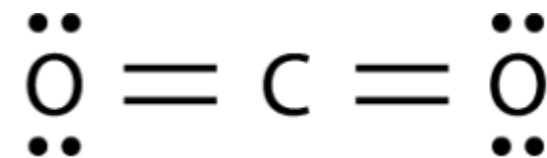
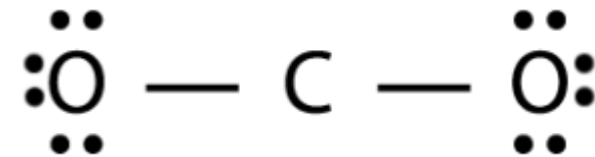
5. Minimer den formelle ladning (FM)* eller i simple tilfælde brug oktatreglen (kun elementer i 1. og 2. række i det periodiske system)

$$\text{FM}(\text{C}) = 4 - 0 - \frac{1}{2} \cdot 4 = 2$$

$$\text{FM}(\text{O}) = 6 - 6 - \frac{1}{2} \cdot 2 = -1$$

$$\text{FM}(\text{C}) = 4 - 0 - \frac{1}{2} \cdot 8 = 0$$

$$\text{FM}(\text{O}) = 6 - 4 - \frac{1}{2} \cdot 4 = 0$$

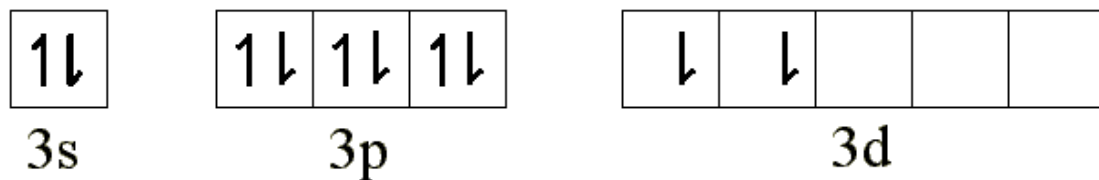
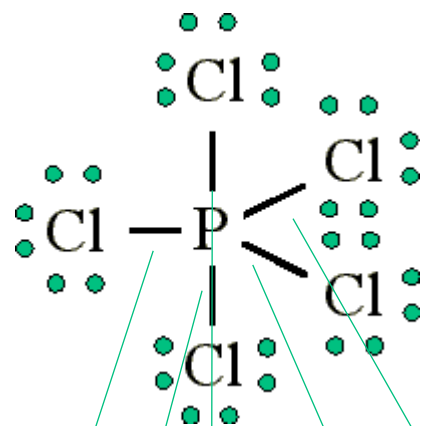


*FM = (valens elektroner af frie atom) - (lone pair elektroner) -
0.5(delte elektroner)

Lewis struktur – fravigelser fra oktet-reglen

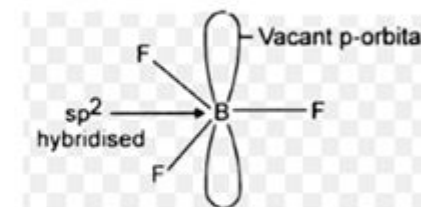
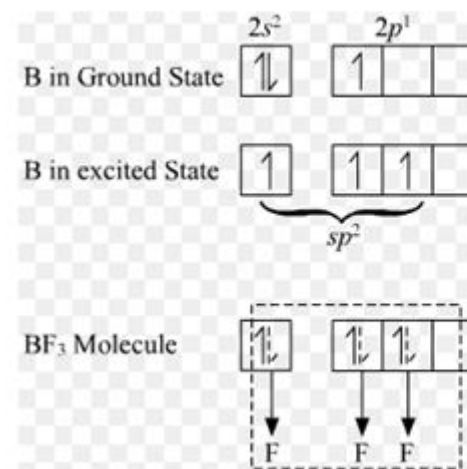
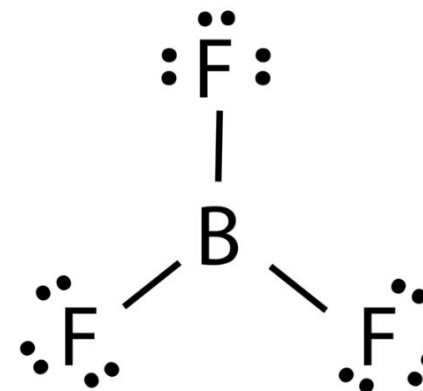
Flere end otte e^- :

- Typisk for $Z > 14$



Mindre end otte e^- :

- Typisk for 13 gruppe (B, Al, Ga)

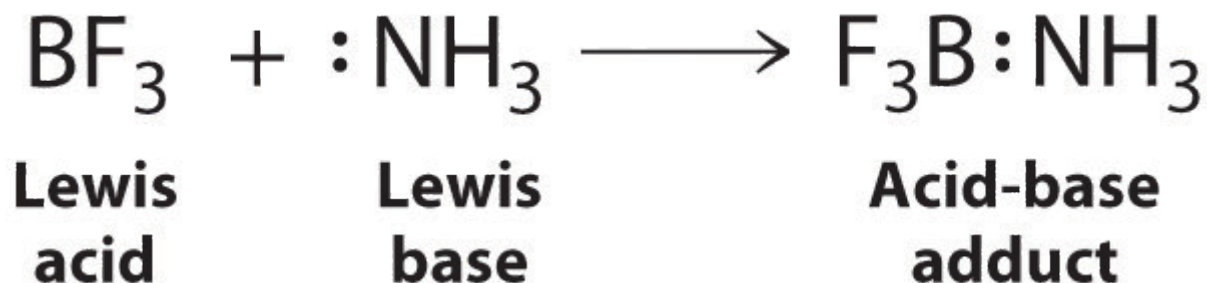


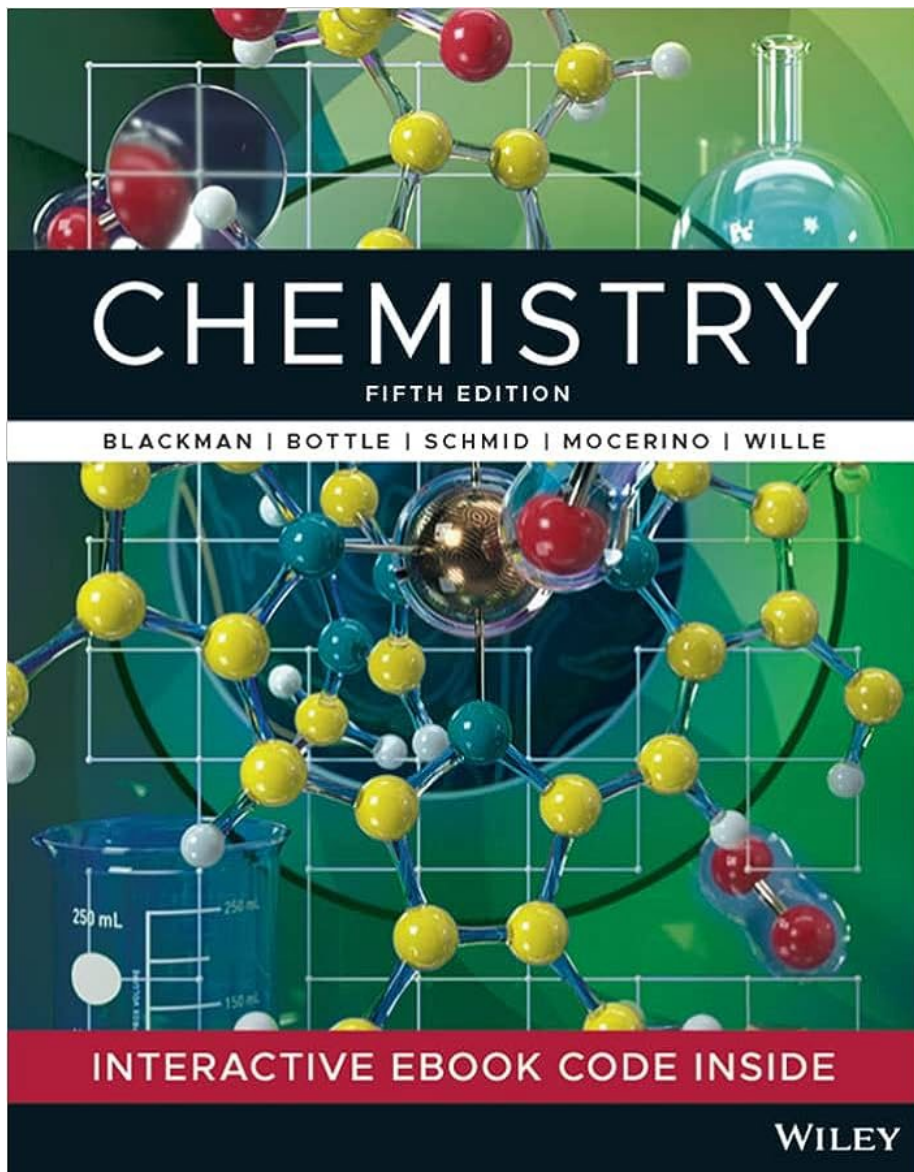
Lewis syre base (detour)

- Lewis syre kan modtage et elektronpar – elektrofil
- Lewis base kan donere et elektronpar – nukleofil

Defenition – Brønsted-Lowry

- Syre: Et stof, der kan afgive en proton (H^+)
 $H_2O(l) + HCl(aq) \rightarrow H_3O^+(aq) + Cl^-(aq)$
- Base: Et stof, der kan optage en proton (H^+)
 $H_2O(aq) + NH_3(g) \rightarrow OH^-(aq) + NH_4^+(aq)$





KE559

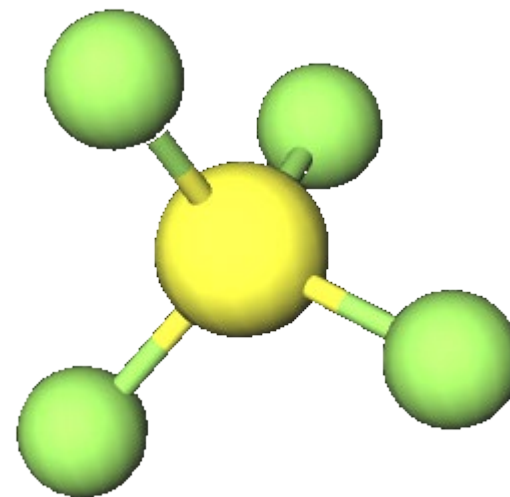
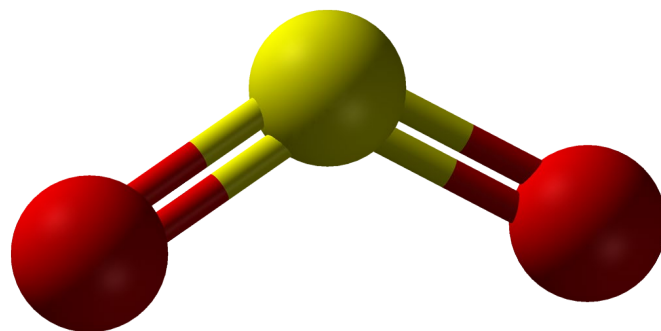
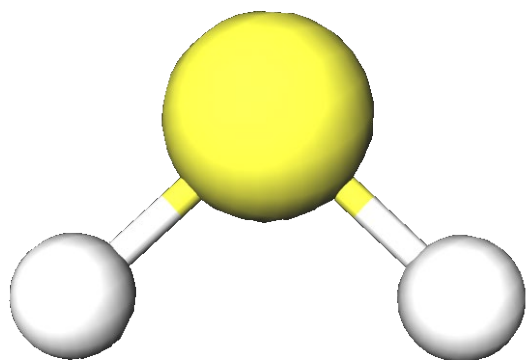
Kemisk binding VSEPR modellen

Blackman 5th edition

p. 197-236

Valence-shell-electron-pair-repulsion (VSEPR)

Metode til at forudsige geometrisk form af molekyler



Valence-shell-electron-pair-repulsion (VSEPR)

1. Tegn Lewis struktur
2. Tæl par af elektroner (både bindinger og lone pairs) omkring atomet
=> forudsig umiddelbar geometri
3. Tag højde for størrelsen af frastødning mellem elektronparene

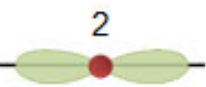
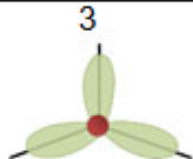
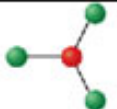
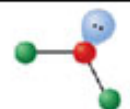
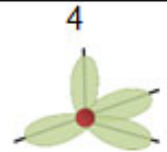
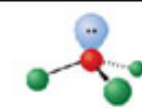
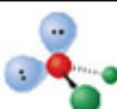
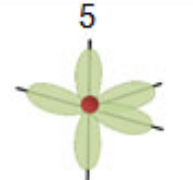
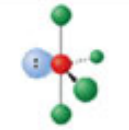
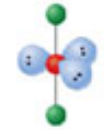
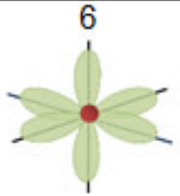
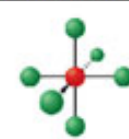
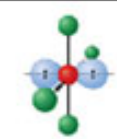
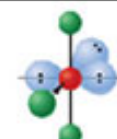
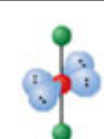
Antal elektronpar	Geometri
2	Lineær
3	Trigonalt plant
4	Tetrahedralt
5	Trigonal bipyramidalt
6	Oktahedralt

$LP - LP > BP - LP > BP - BP$

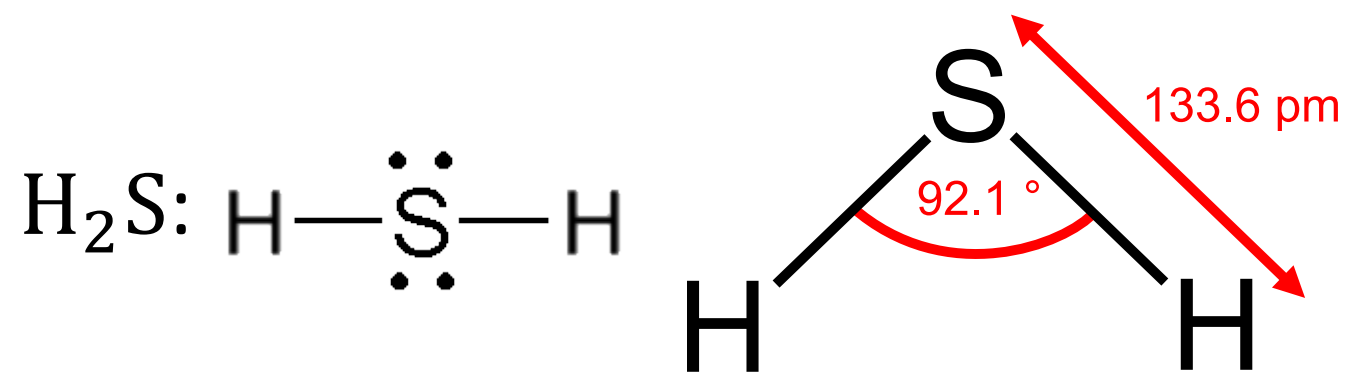
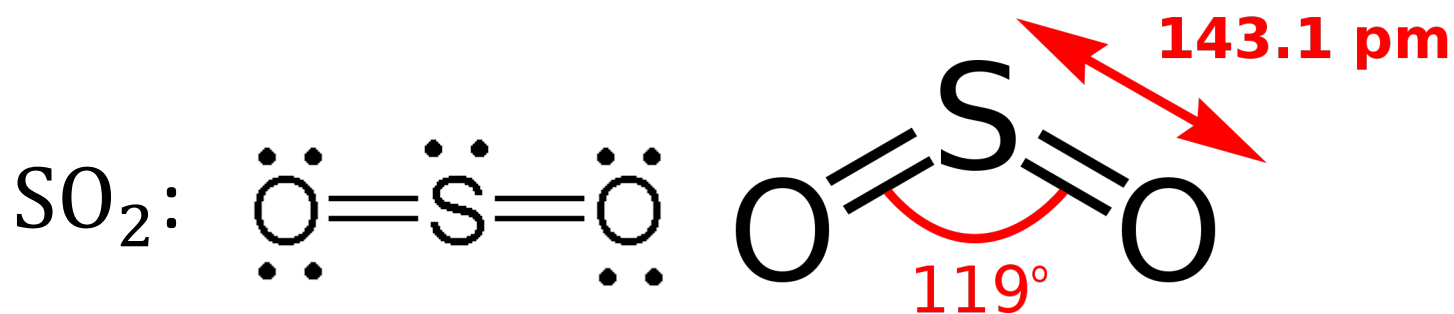
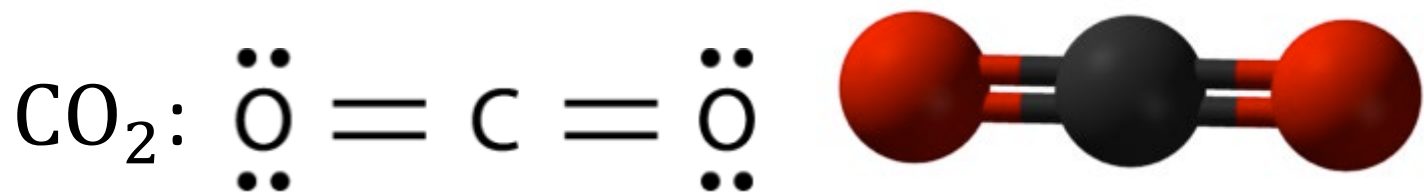
LP – Lone pairs

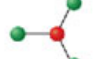
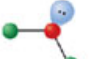
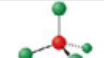
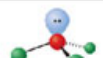
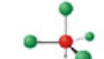
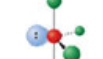
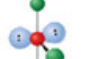
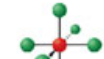
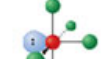
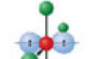
BP – Bonding pairs

VSEPR

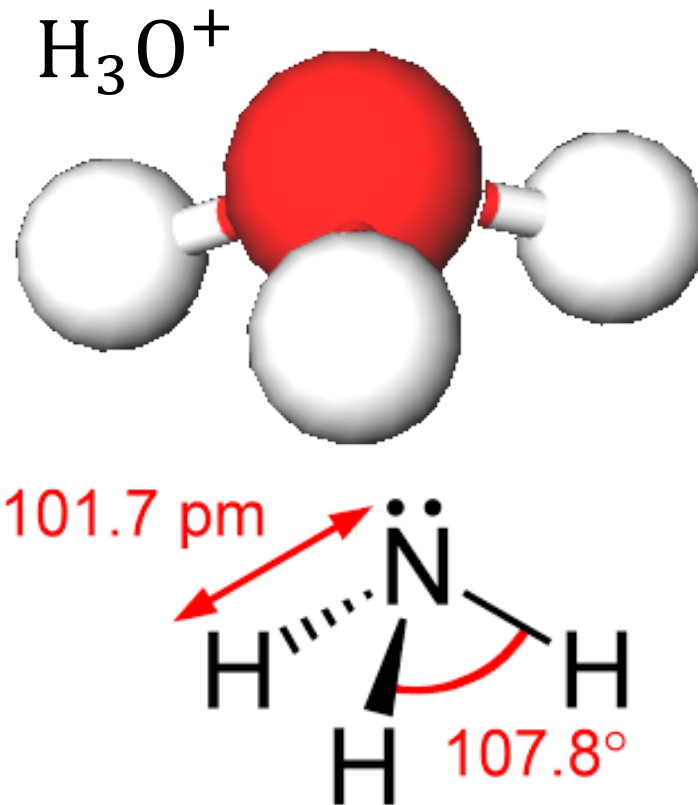
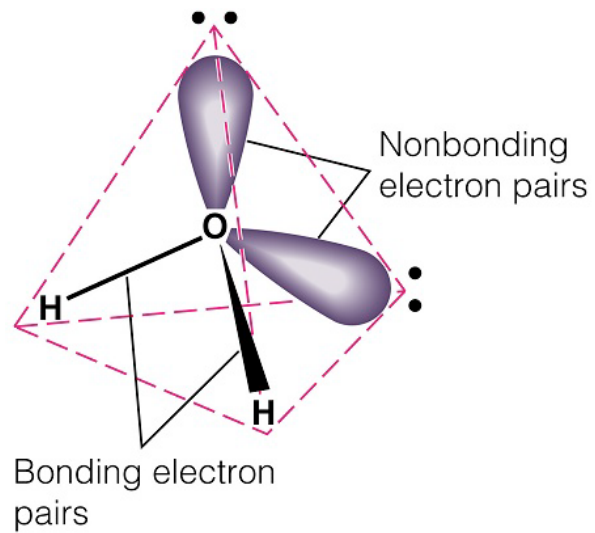
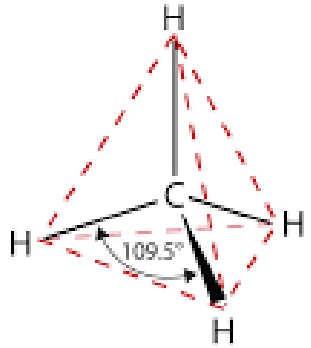
Number of Electron Dense Areas	Electron- Pair Geometry	Molecular Geometry				
		No Lone Pairs	1 lone Pair	2 lone Pairs	3 lone Pairs	4 lone Pairs
2 	Linear	 Linear				
3 	Trigonal planar	 Trigonal planar	 Bent			
4 	Tetrahedral	 Tetrahedral	 Trigonal pyramidal	 Bent		
5 	Trigonal bipyramidal	 Trigonal bipyramidal	 Sawhorse	 T-shaped	 Linear	
6 	Octahedral	 Octahedral	 Square pyramidal	 Square planar	 T-shaped	 Linear

VSEPR -Eksempler



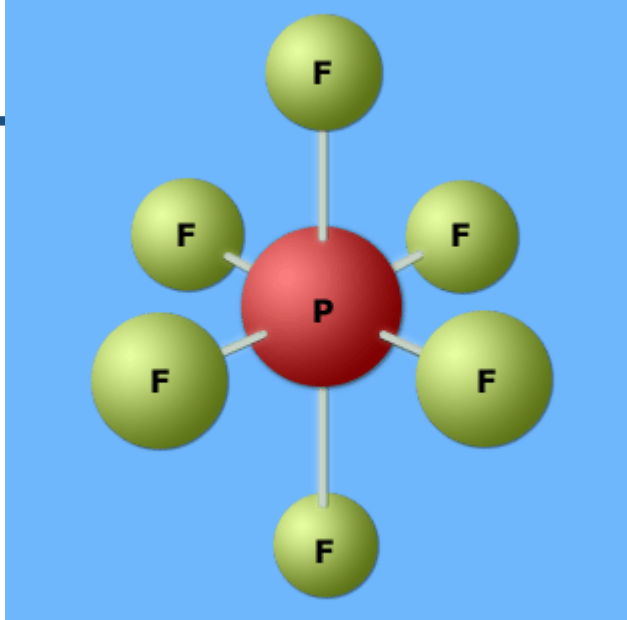
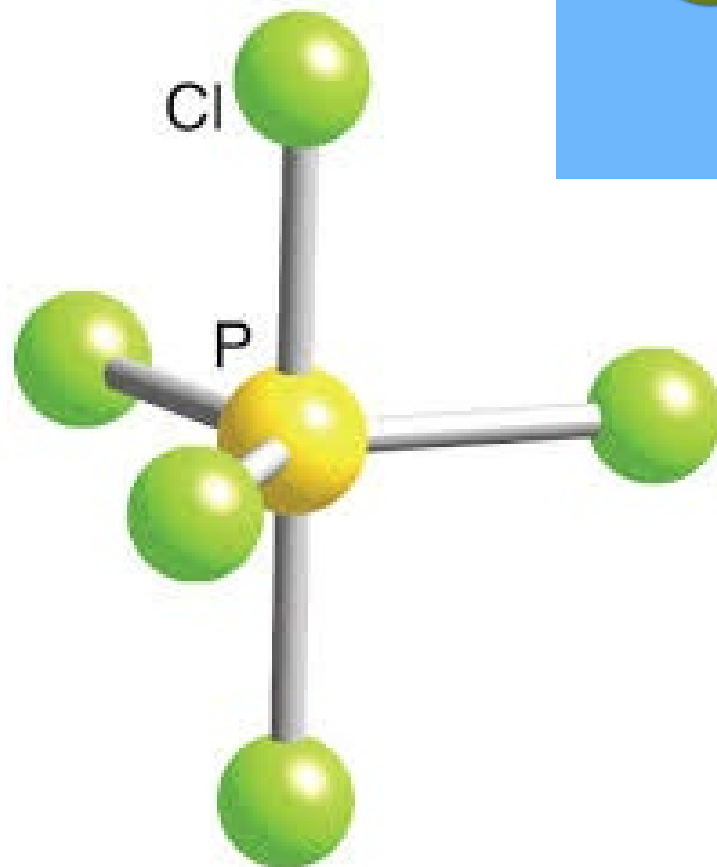
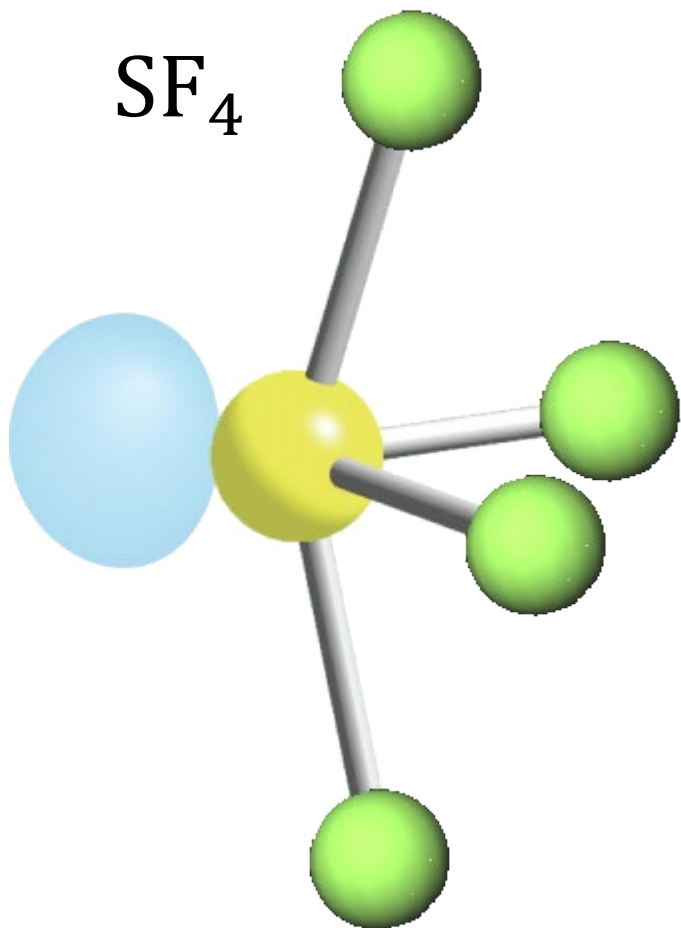
Number of Electron Dense Areas	Electron-Pair Geometry	Molecular Geometry				
		No Lone Pairs	1 lone Pair	2 lone Pairs	3 lone Pairs	4 lone Pairs
2	Linear					
3	Trigonal planar					
4	Tetrahedral					
5	Trigonal bipyramidal					
6	Octahedral					

VSEPR -Eksempler



Number of Electron Dense Areas	Electron- Pair Geometry	Molecular Geometry				
		No Lone Pairs	1 lone Pair	2 lone Pairs	3 lone Pairs	4 lone Pairs
2	Linear	Linear				
3	Trigonal planar	Trigonal planar	Bent			
4	Tetrahedral	Tetrahedral	Trigonal pyramidal	Bent		
5	Trigonal bipyramidal	Trigonal bipyramidal	Sawhorse	T-shaped	Linear	
6	Octahedral	Octahedral	Square pyramidal	Square planar	T-shaped	Linear

VSEPR - Eksempler



Molecular Geometry						
			0 Lone Pairs	1 lone Pair	2 lone Pairs	3 lone Pairs
2	Linear	Linear				
3	Trigonal planar	Trigonal planar	Bent			
4	Tetrahedral	Tetrahedral	Trigonal pyramidal	Bent		
5	Trigonal bipyramidal	Trigonal bipyramidal	Sawhorse	T-shaped	Linear	
6	Octahedral	Octahedral	Square pyramidal	Square planar	T-shaped	Linear

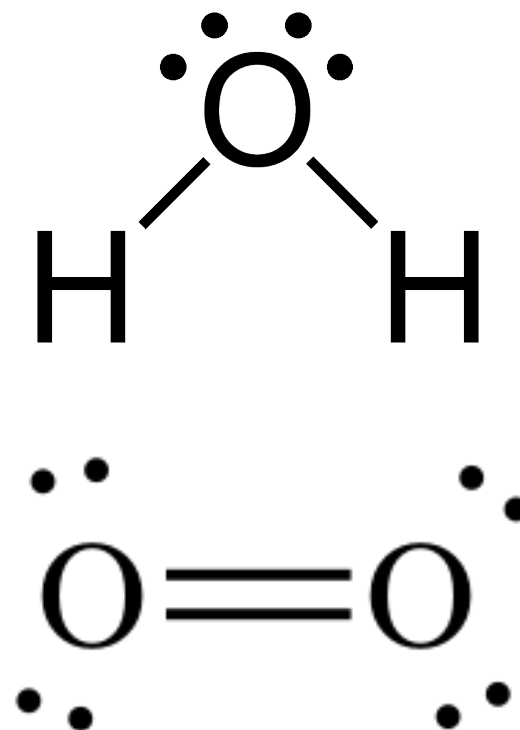
Opgave

Tegn Lewis-struktur for trioxygen (eller ozon) O_3 . Vil der være nogen resonansformer? Vil ozonmolekylet være lineært eller bøjet?

Lewis struktur - fremgangsmåde

Fremgangsmåde (side 205)

1. Tæl hvor mange valenselektroner hele molekylet har (husk at tage højde for evt. ladning)
2. Tegn molekylet med kun enkeltbindinger
3. Placer resten af valens elektronerne ved at placere tre lone-pairs på hvert af de ydre atomer (dog ikke på H)
4. Placer de resterende valens elektroner som lone pairs på det centrale atom
5. Minimer den formelle ladning på alle atomer



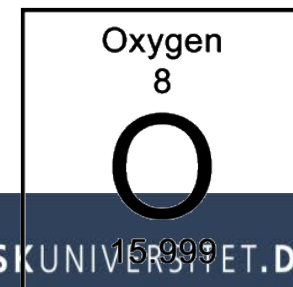
Fremgangsmåde (side 205)

1. Tæl hvor mange valenselektroner hele molekylet har (husk at tage højde for evt. ladning)

$3 * 6 = 18$ – elektronerne i den fyldte 1S orbital tages ikke med

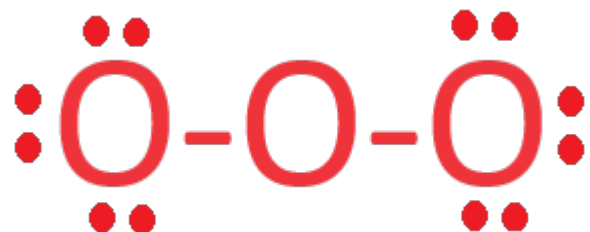
2. Tegn molekylet med kun enkeltbindinger

O-O-O – dvs vi bruger 4 valens-elektroner



Lewis struktur - fremgangsmåde

1. Placer resten af valens elektroner ved at placere tre lone-pairs på hvert af de ydre atomer (dog ikke på H)



Brugt; $4 + 2 \cdot 6 = 16$

2. Placer de resterende valens elektroner som lone pairs på det centrale atom



5. Minimer den formelle ladning (FM)*

$$\text{FM}(\text{O1}) = 6 - 6 - \frac{1}{2} \cdot 2 = -1$$

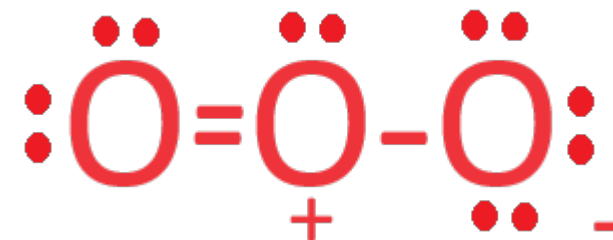
$$\text{FM}(\text{O2}) = 6 - 2 - \frac{1}{2} \cdot 4 = +2$$

$$\text{FM}(\text{O3}) = 6 - 6 - \frac{1}{2} \cdot 2 = -1$$

$$\text{FM}(\text{O1}) = 6 - 4 - \frac{1}{2} \cdot 4 = 0$$

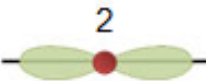
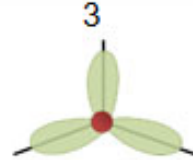
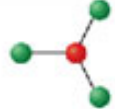
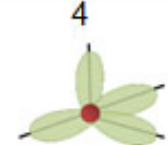
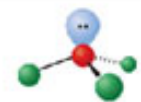
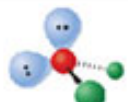
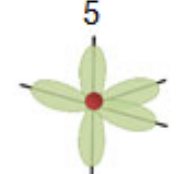
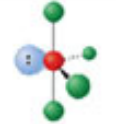
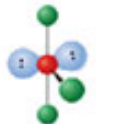
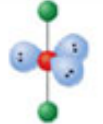
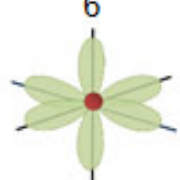
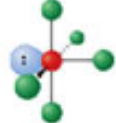
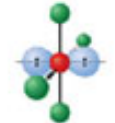
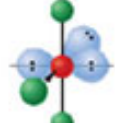
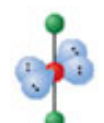
$$\text{FM}(\text{O2}) = 6 - 2 - \frac{1}{2} \cdot 6 = +1$$

$$\text{FM}(\text{O3}) = 6 - 6 - \frac{1}{2} \cdot 2 = -1$$



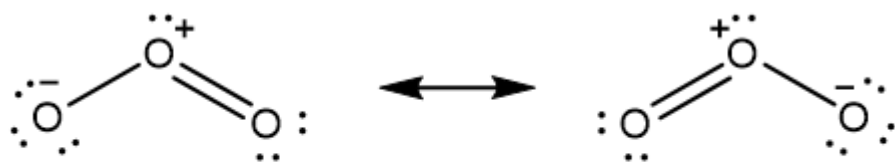
*FM = (valens elektroner af frie atom) - (lone pair elektroner) -
0.5(delte elektroner)

VSEPR

Number of Electron Dense Areas	Electron- Pair Geometry	Molecular Geometry				
		No Lone Pairs	1 lone Pair	2 lone Pairs	3 lone Pairs	4 lone Pairs
	Linear	 Linear				
	Trigonal planar	 Trigonal planar	 Bent			
	Tetrahedral	 Tetrahedral	 Trigonal pyramidal	 Bent		
	Trigonal bipyramidal	 Trigonal bipyramidal	 Sawhorse	 T-shaped	 Linear	
	Octahedral	 Octahedral	 Square pyramidal	 Square planar	 T-shaped	 Linear

Opgave

Tegn Lewis-struktur for trioxygen (eller ozon) O_3 . Vil der være nogen resonansformer? Vil ozonmolekylet være lineært eller bøjet?



Det ses at der er et lone pair på oxygen i midten, hvilket giver anledning til knækket.

Opgave 2

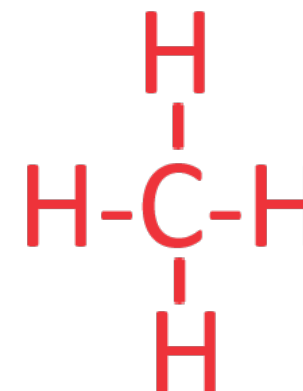
Optegn Lewisstrukturer (inkl. lone pairs og formelle ladninger), der opfylder oktetreglen, for følgende molekyler (sumformlen er opgivet i parentes som hjælp).

a) Methan (CH_4)

b) Methanamin (CH_5N)

Lewis methan (CH₄)

1. Valens elektroner; $4 + (4 \cdot 1) = 8$
2. Tegn molekylet med kun enkeltbindinger
3. Placer resten af valenselektroner på de ydre atomer (undtagen H)
4. Placer resten af valenselektronerne på det centrale elektron
5. Minimer den formelle ladning på alle atomer

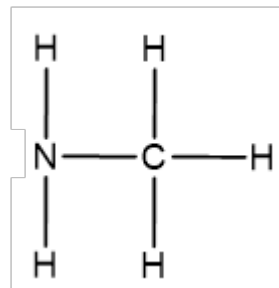


Lewis methanamine (CH₅N)

1. Valens elektroner; $4 + (5 \cdot 1) + 5 = 14$
2. Tegn molekylet med kun enkeltbindinger
3. Placer resten af valenselektroner på de ydre atomer (undtagen H)
4. Placer resten af valenselektronerne på det centrale elektron
5. Minimer den formelle ladning på alle atomer

Lewis methanamine

1. Valens elektroner; $4 + (5 \cdot 1) + 5 = 14$
2. Tegn molekylet med kun enkeltbindinger



3. Placer resten af valenselektroner på de ydre atomer (undtagen H)
4. Placer resten af valenselektronerne på det centrale elektron
5. Minimer den formelle ladning på alle atomer

Hovedkvantetal, n og det periodiske system

Række

1

2

3

4

5

6

7

n

Atomic Number

7

Chemical Symbol

N

Chemical Name

NITROGEN

Atomic Weight

14

NON-METALS

METALS

1 H HYDROGEN 1																	2 He HELIUM 4				
3 Li LITHIUM 7	4 Be BERYLLIUM 9															5 B BORON 11	6 C CARBON 12	7 N NITROGEN 14	8 O OXYGEN 16	9 F FLUORINE 19	10 Ne NEON 20
11 Na SODIUM 23	12 Mg MAGNESIUM 24															13 Al ALUMINUM 27	14 Si SILICON 28	15 P PHOSPHORUS 31	16 S SULFUR 32	17 Cl CHLORINE 35	18 Ar ARGON 40
19 K POTASSIUM 39	20 Ca CALCIUM 40	21 Sc SCANDIUM 45	22 Ti TITANIUM 48	23 V VANADIUM 51	24 Cr CHROMIUM 52	25 Mn MANGANESE 55	26 Fe IRON 56	27 Co COBALT 59	28 Ni NICKEL 59	29 Cu COPPER 64	30 Zn ZINC 65	31 Ga GALLIUM 70	32 Ge GERMANIUM 73	33 As ARSENIC 75	34 Se SELENIUM 79	35 Br BROMINE 80	36 Kr KRYPTON 84				
37 Rb RUBIDIUM 85	38 Sr STRONTIUM 88	39 Y YTTRIUM 89	40 Zr ZIRCONIUM 91	41 Nb NIOBIUM 93	42 Mo MOLYBDENUM 96	43 Tc TECHNETIUM 98	44 Ru RUTHENIUM 101	45 Rh RHODIUM 103	46 Pd PALLADIUM 106	47 Ag SILVER 108	48 Cd CADMIUM 112	49 In INDIUM 115	50 Sn TIN 119	51 Sb ANTIMONY 122	52 Te TELLURIUM 128	53 I IODINE 127	54 Xe XENON 131				
55 Cs CESIUM 133	56 Ba BARIUM 137			72 Hf HAFNIUM 178	73 Ta TANTALUM 181	74 W TUNGSTEN 184	75 Re RHENIUM 186	76 Os OSMIUM 190	77 Ir IRIDIUM 192	78 Pt PLATINUM 195	79 Au GOLD 197	80 Hg MERCURY 201	81 Tl THALLIUM 204	82 Pb LEAD 207	83 Bi BISMUTH 209	84 Po POLONIUM 209	85 At ASTATINE 210	86 Rn RADON 222			
87 Fr FRANCIUM 223	88 Ra RADIUM 226			104 Rf RUTHERFORDIUM 267	105 Db DUBNIUM 268	106 Sg SEABORGIUM 271	107 Bh BOHRERIUM 272	108 Hs HASSIUM 277	109 Mt MEITNERIUM 276	110 Ds DARMSTADTIUM 281	111 Rg ROENTGENIUM 280	112 Cp COPERNICIUM 285	113 Uut UNUNTRIUM 284	114 Uuq UNUNQUADIUM 289	115 Uup UNUNPENTIUM 288	116 Uuh UNUNHEXIUM 291	117 Uus UNUNSEPTIUM not yet observed	118 Uuo UNUNOCTIUM 294			

KEY

SOLID at room temp

LIQUID at room temp

GAS at room temp

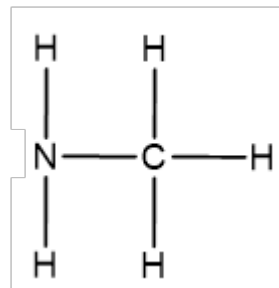
RADIOACTIVE

Artificially created

57 La LANTHANUM 139	58 Ce CERIUM 140	59 Pr PRASEODYMIUM 141	60 Nd NEODYMIUM 144	61 Pm PROMETHIUM 145	62 Sm SAMARIUM 150	63 Eu EUROPIUM 152	64 Gd GADOLINIUM 157	65 Tb TERBIUM 159	66 Dy DYSPROSIUM 163	67 Ho HOLMIUM 165	68 Er ERBIUM 167	69 Tm THULIUM 169	70 Yb YTTERBIUM 173	71 Lu LUTETIUM 175
89 Ac ACTINIUM 227	90 Th THORIUM 232	91 Pa PROTACTINIUM 231	92 U URANIUM 238	93 Np NEPTUNIUM 237	94 Pu PLUTONIUM 244	95 Am AMERICIUM 243	96 Cm CURIUM 247	97 Bk BERKELIUM 247	98 Cf CALIFORNIUM 251	99 Es ENSTEINIUM 252	100 Fm FERMIUM 257	101 Md MENDELEVIUM 258	102 No NOBELIUM 259	103 Lr LAWRENCIUM 262

Lewis methanamine

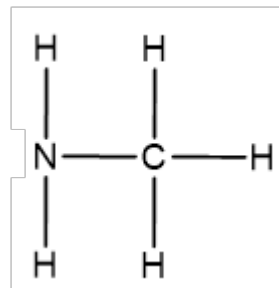
1. Valens elektroner; $4 + (5 \cdot 1) + 5 = 14$
2. Tegn molekylet med kun enkeltbindinger



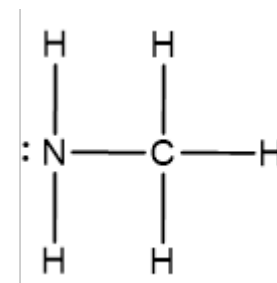
3. Placer resten af valenselektroner på de ydre atomer (undtagen H)
4. Placer resten af valenselektronerne på det centrale elektron
5. Minimer den formelle ladning på alle atomer

Lewis methanamine

1. Valens elektroner; $4 + (5 \cdot 1) + 5 = 14$
2. Tegn molekylet med kun enkeltbindinger



3. Placer resten af valenselektroner på de ydre atomer (undtagen H)



4. Placer resten af valenselektronerne på det centrale elektron
5. Minimer den formelle ladning på alle atomer

Hvad har vi snakket om ?

- Bindinger og elektronegativitet
- Bindingstyper
 - Kovalent – deler elektroner ligeligt
 - Polær kovalent – deler elektroner men ikke ligeligt
 - Ionbinding – elektroner er overført fra et atom til et andet
- Lewis-prik-strukturer
- VSEPR-modellen

Næste gang

- Næste gang starter vi med at snakke om termodynamik
 - Læs kapitel 8, side 347-378
- Dette vil gennemsnitligt tage jer 6 timer